

Lecture Notes on

Quantum Computing and Cryptography

Part I – Linear Algebra and Quantum Mechanics

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Dr Marco Paviotti
Department of Computing
University of Kent



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1 Introduction

Quantum computing is a paradigm which uses the principles of quantum mechanics to process information. Unlike classical computers, which operate on bits that represent either 0 or 1, quantum computers use quantum bits, or *qubits*. Qubits can exist in a superposition of states, represented mathematically as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers that determine the probabilities of the qubit being measured as 0 or 1. This property allows quantum computers to explore multiple solutions simultaneously, offering a significant advantage for certain computational tasks.

Another key feature of quantum systems is *entanglement*, where qubits become correlated in such a way that the state of one qubit depends on the state of another, regardless of the physical distance between them. This enables highly coordinated operations across qubits, a phenomenon that classical systems cannot replicate. To perform computations, quantum computers use *quantum gates*, such as the Hadamard gate, which creates superposition, and the CNOT gate, which introduces entanglement. These gates manipulate qubits to execute algorithms in fundamentally different ways from classical logic gates.

Quantum computing holds promise for solving problems that are intractable for classical computers. For instance, Shor's algorithm can factorize large numbers exponentially faster than classical methods, posing a challenge to current cryptographic systems. Grover's algorithm, on the other hand, provides a quadratic speedup for searching unsorted databases. Quantum computers are also uniquely suited for simulating quantum systems, making them invaluable for advancing material science, drug discovery, and fundamental physics.

Despite its potential, quantum computing faces significant challenges. Qubits are highly sensitive to their environment, leading to *decoherence*, where quantum states lose their coherence over time. Additionally, quantum systems are prone to errors, necessitating robust error correction methods. Scaling quantum computers to include thousands or millions of qubits also remains a formidable engineering hurdle.

As research and technology progress, quantum computing promises to revolutionize fields such as cryptography, artificial intelligence, and healthcare, offering transformative solutions to some of humanity's most complex problems.

In these notes we will give the reader a comprehensive introduction to how quantum computing works. First we will introduce the reader to basic mathematical concepts like set theory and linear algebra, which are essential to understand how quantum mechanics work.

1.1 Further Reading

Most linear algebra books cover all the materials from Section 2. I have drawn most material from

Elementary Linear Algebra: Applications Version, Howard Anton and Chris Rorres, Tenth Edition [1].

Here's the chapters that are relevant for this course:

- Chapter 1. Sections: 1.3, 1.4
- Chapter 3. Sections: 3.1, 3.2, 3.3, 3.5
- Chapter 4. Sections: 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.9, 4.10
- Chapter 5. Sections: 5.1, 5.2, 5.3
- Chapter 6. Sections: 6.1, 6.2, 6.3
- Chapter 7. Sections: 7.1, 7.2, 7.5

For Section 3 the source is

Quantum Computation and Quantum Information, M. Nielsen, and I. Chuang. Cambridge University Press, (2000) [3].

Here's the chapters that are relevant for this course:

- Chapter 2. Section 2.1. Linear Algebra.
 - Sections: 2.1.1, 2.1.2, 2.1.3, 2.1.4, 2.1.5, 2.1.6, 2.1.7, 2.1.8, 2.1.10
- Chapter 2. Section 2.2. Postulates of Quantum Mechanics.
 - Sections: 2.2.1, 2.2.2, 2.2.3, 2.2.4, 2.2.5, 2.2.6, 2.2.7, 2.2.8
- Chapter 2. Section 2.4

If you need a refresher on basic logic, set theory and trigonometry read Appendix A and B.

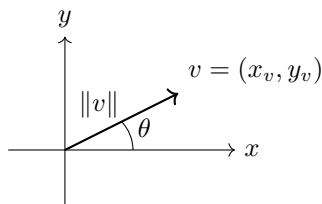


Figure 1: A vector

2 Linear Algebra

Linear algebra is a branch of mathematics that deals with vector spaces, linear equations, and transformations. It is foundational to many areas of mathematics and has applications in physics, engineering, economics, but in particular, computer science, machine learning, quantum computing and numerical analysis.

The purpose of linear algebra is to study the structure and behavior of vectors, matrices, and systems of linear equations. This is useful in quantum because qbits will be represented by probability vectors and quantum algorithms will be represented essentially by compositions of linear transformations or matrices.

2.1 Vector Spaces

A vector space is a set of objects called *vectors*. In the case of the plane $\mathbb{R}^2$ a vector v is simply a pair of coordinates (x_v, y_v) on the plane. Equivalently, a vector in $\mathbb{R}^2$ can be seen as an arrow starting from the center to the point (x_v, y_v) as in Figure 1. This arrow induces a length, called the *modulus* of the vector, and an angle θ between the x -axis and the vector itself.

If we think of a vector as representing a *change*, such as movement, force, or transformation. Then adding two vectors together means performing one change, then perform the other. In other words, *addition combines contributions into a single result*. The addition operation is given by point-wise addition. For example, adding $(1, 2)$ and $(3, 4)$ gives $(1 + 3, 4 + 2)$, while multiplying $(1, 2)$ by a scalar 3 results in $(3, 6)$. An example of addition of two vectors is depicted in Figure 2a.

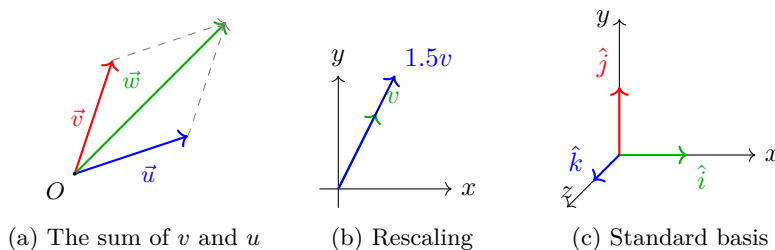
A vector can also be stretched, shrunk, or flipped. If the vector is seen as a force, this is equivalent to “tuning” their strength. The operation associated with this idea is called *scalar multiplication* (Figure 2b). A *scalar* is a number and can be used to adjust the *magnitude* and possibly the *direction* of a vector. For example, if a vector represents the speed of a certain mass, then multiplying it by 2 results in doubling the vector doubles the speed. Mathematically, given a scalar $c \in \mathbb{R}$ and a vector $(x_v, y_v) \in \mathbb{R}^2$ scalar multiplication is pointwise multiplication $(c \cdot x_v, c \cdot y_v)$.

Similarly, in three-dimensional Euclidean space $\mathbb{R}^3$, vectors are represented as triples (x_v, y_v, z_v) . Given two vectors $v = (x_v, y_v, z_v)$ and $w = (x_w, y_w, z_w)$ the addition yields $v + w = (x_v + x_w, y_v + y_w, z_v + z_w)$ and scalar multiplication of v by a real number $c \in \mathbb{R}$ gives $cv = (cx_v, cy_v, cz_v)$.

These examples lead to the general definition of vector space.

Definition 2.1 (Vector Space). *A vector space over a set of scalars $\mathbb{F}$ is a set with a binary operation associated with a neutral element $(V, +, 0)$ together with a scalar multiplication $\cdot : \mathbb{F} \times V \rightarrow V$. These operations must satisfy the following axioms for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$:*

1. $v + 0 = v$
2. $(u + v) + w = u + (v + w)$.
3. $u + v = v + u$.
4. For each $v \in V$, there exists a vector (called the inverse) $-v \in V$ such that $v + (-v) = 0$.



5. $1v = v$ for all $v \in V$ and $1 \in \mathbb{F}$

6. $a(u + v) = au + av$

7. $(a + b)v = av + bv$

8. $a(bv) = (ab)v$

Most examples of vector spaces in this section will be over real numbers $\mathbb{R}$. However, later in Section 3 we will start using vector spaces over the complex numbers, written $\mathbb{C}$. This is simply because vector spaces over the complexes have all the properties we need to model quantum computations and all the examples that work for the reals, work also for the complexes since $\mathbb{R} \subseteq \mathbb{C}$, by sending a real number r into a complex numbers $r + i0$ with the null imaginary part.

2.2 Linear Transformations

Now that we introduced vectors we are going to show how these can be transformed in the space by means of a transformation. First consider the special vectors $\hat{i} = (1, 0, 0)$ and $\hat{j} = (0, 1, 0)$ and $\hat{k} = (0, 0, 1)$ in $\mathbb{R}^3$. We call these vectors the *standard basis vectors* (Figure 2c) and we will see shortly that these are very important vectors in $\mathbb{R}^3$. For the sake of the following examples, we will only use two dimensions and thus we will only need $\hat{i}$ and $\hat{j}$.

Now the basic trick to construct a transformation in 2 dimensions is to look at the basis vectors and figure out where we want these vectors to go after the transformation is applied. Then the transformation is obtained by constructing a 2×2 matrix where the first column is where $\hat{i}$ has to land and the second column is the vector where $\hat{j}$ has to land.

2.2.1 Rotations

Assume that we want to rotate $\hat{i}$ and $\hat{j}$ to the left by 90° . This means that $\hat{i}$ has to land in the point in space with coordinates $(0, 1)$ and $\hat{j}$ has to land where $\hat{i}$ was before applying the transformation, that is $(-1, 0)$. In order to achieve this, we construct a 2×2 matrix where the first column is where $\hat{i}$ lands and the second column is where $\hat{j}$ lands. We call this matrix R_{90} which stands for “rotation by 90° ”:

$$R_{90} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

At this point if we wanted to rotate any vector $v \in \mathbb{R}^2$ by 90° anti-clockwise we just have to compute the result of applying the matrix R_{90} to the vector v . This is done by computing matrix-vector multiplication, written $R_{90}v$. The result of applying R_{90} to $\hat{i}$ and $\hat{j}$ is depicted in Figure 4a.

It is easy to convince yourself that any other vector in space is rotated by 90° degrees anti-clockwise accordingly.

Exercise 2.1. Compute $R_{90}v$ for v being the vector $(1, 2)$, $(3, 1)$ and $(2, 1)$.

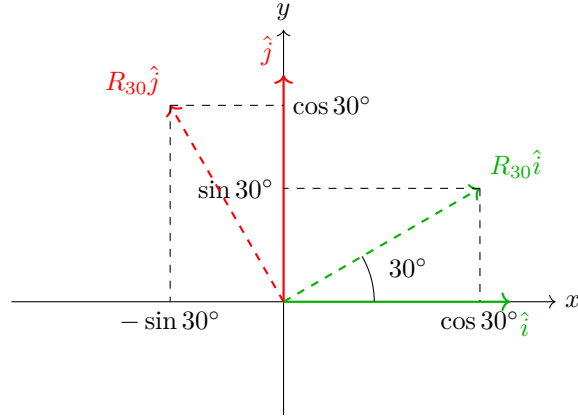


Figure 3: Rotation by -30°

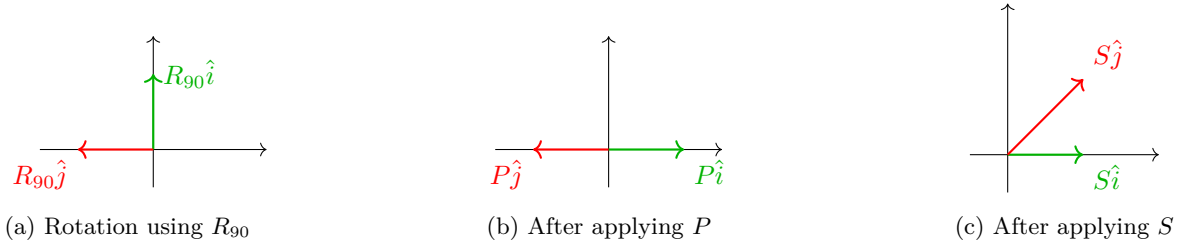


Figure 4: Linear transformations on the standard basis

In general, we can rotate the basis vectors by an arbitrary angle θ by computing the sine and cosine of the angle. The matrix representing this transformation is defined as follows:

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Here $(\cos \theta, \sin \theta)$ is where $\hat{i}$ has to land and $(-\sin \theta, \cos \theta)$ is where $\hat{j}$ has to land. To convince the reader this is the right thing to do consider when $\theta = 30^\circ$ and take a look at Figure 3 which depicts a rotation of 30 degrees anti-clockwise. The coordinates of the vector $R_{30}\hat{i}$, for example, are simply given by taking the cosine of the angle 30° and the sine of the same angle and similarly, for the vector $R_{30}\hat{j}$.

Exercise 2.2. Compute the rotation matrix for the angle $\theta = 45^\circ$.

2.2.2 Projections

In the same way we constructed the rotation matrix we are now going to construct a matrix which maintains the vectors on the real line (those which have 0 in the y component) where they are and squish all the other vectors into the real line as well.

In order to do this we look at the basis vectors $\hat{i}$ and $\hat{j}$ again and think where we want these to land after the transformation is applied. The first column is the one where we expect $\hat{i}$ to land, so it is going to be $(1, 0)$ which is $\hat{i}$ itself. The second column is where we want $\hat{j}$ to land. We have many options to bring $\hat{j}$ down to the real line. We choose $(-1, 0)$. Thus the projection matrix associated with this transformation is defined by:

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

The effect of applying P to the standard basis vectors is depicted in Figure 4b.

2.2.3 Shear

Next up we are going to show how to construct the shear transformation. The shear matrix tilts vectors who have a non-zero y -component. This is similar to the projection matrix in that the vectors that lie on the x -axis will stand still while all the others will tilt accordingly.

The objective is to tilt and stretch $\hat{j}$ on the right by 45° degrees, thus landing in position $(1, 1)$. The shear matrix is thus give by:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The reader should convince themselves that S leaves the x -axis invariant, that is $S\hat{i} = \hat{i}$, while the vector $\hat{j}$ is mapped to $S\hat{j} = (1, 1)$. More generally, a vector (x, y) is transformed to

$$S(x, y) = (x + y, y)$$

which means that each point is shifted horizontally by an amount proportional to its height y , while its vertical coordinate remains unchanged. Thus horizontal lines stay horizontal, the origin is fixed, areas are preserved and the transformation produces a slanted, or “sheared,” version of the plane. The effect of the shear applied to the basis vectors of $\mathbb{R}^2$ is depicted in Figure 4c.

2.2.4 Linear Transformations

Whenever we have a matrix, the operation of computing matrix-vector multiplication extends to a function between vector spaces. For example, assume we have a matrix $M \in \mathbb{R}^{n \times m}$. Now given a vector $v \in \mathbb{R}^m$, we can use matrix-vector multiplication to transform this vector into a vector $Mv \in \mathbb{R}^n$. Hence, the matrix M can be thought as a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ sending a vector in the vector space $\mathbb{R}^m$ into the vector space $\mathbb{R}^n$. If this transformation respects the operations of addition and scalar multiplication, as we shall see shortly, then this transformation is additionally linear. All the matrices are examples of linear transformations.

Definition 2.2 (Linear Transformation). *A linear map is a function $f : V \rightarrow W$ between vector spaces, with the following properties, for all $v, w \in V$ and $s \in \mathbb{C}$:*

$$f(v + w) = f(v) + f(w) \tag{1}$$

$$f(s \cdot v) = s \cdot f(v) \tag{2}$$

An anti-linear map is a function that satisfies (1) but instead of (2), satisfies

$$f(s \cdot v) = s^* \cdot f(v)$$

where s^ is the complex conjugate.*

These two properties ensure that a linear transformation preserves the structure of the vector space. A simple example of a linear transformation is a scaling transformation. If f is a scaling transformation by a factor of k , then $f(v) = kv$, where k is a scalar. This transformation scales all vectors by the same factor k , preserving their direction but altering their magnitude.

Obviously, when the vector space is over the reals, the linear map is the same as the anti-linear map as the complex conjugate of a real number is the identity.

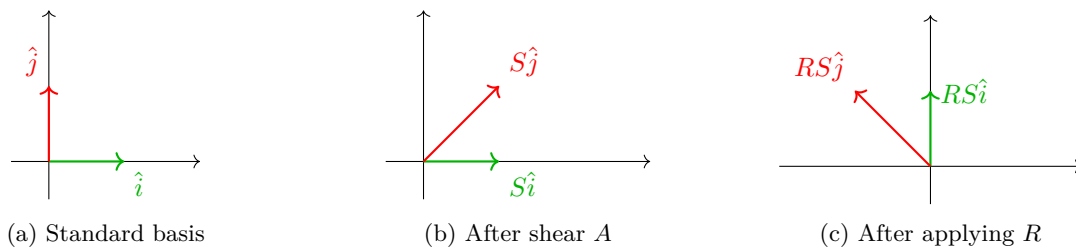


Figure 5: Illustration of the composition $R \circ A$ on the standard basis

2.2.5 Composition

The *composition* of two linear transformations A and B , written $B \circ A$, is the transformation that applies A first and then B to the result is represented by the matrix product BA , because

$$(B \circ A)(v) = B(A(v)) = B(Av) = (BA)v.$$

There is an identity transformation given by the identity matrix I defined as $I_{i,i} = 1$ and $I_{i,j} = 0$ for $i \neq j$ which satisfies

$$A \circ I = I \circ A = A$$

and, moreover, composition is associative, that is

$$A \circ (B \circ C) = (A \circ B) \circ C.$$

However, it is important to notice that composition is *not commutative*. To see this consider a horizontal shear S followed by a rotation by 90° counterclockwise R . The composition *rotation after shear* is RS , which generally produces a different result than *shear after rotation* SR . This is illustrated in Figure 5.

2.3 Determinant

For reasons that are going to be clear in the next sections it is interesting to look at the area of the rectangle formed by the standard basis as in Figure 6a. When a transformation is applied this rectangle can become larger or smaller depending on the transformation. It is therefore useful to understand what is the enlargement factor of the transformation with respect to the standard basis vectors.

This fact is called the *determinant* of a matrix. To understand how this is computed take a look at Figure 6b. This is the picture that is obtained after a certain transformation is applied. In this case, v and w are the vectors obtained by applying the matrix

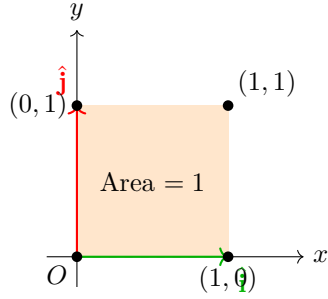
$$A = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$$

to the vectors $\hat{i}$ and $\hat{j}$ respectively. The determinant is simply the area of the parallelogram induced by these two vectors, which is computed simply by computing the area of the whole rectangle from the center to the top-right point $(5, 3)$ and subtracting the leftover areas. In particular, let $a = 4$, $b = 1$, $c = 1$ and $d = 2$, then

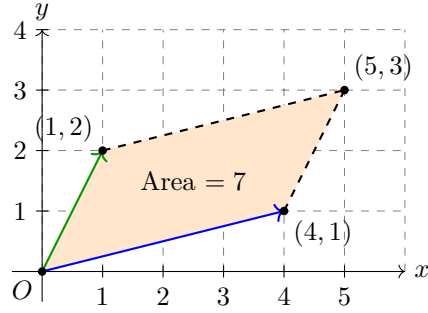
$$\det(A) = (a + b)(d + c) - 2bc - bd/2 - 2(ac/2) = ad - bc$$

In other words, the determinant of a 2×2 matrix is computed by multiplying the elements of the diagonals together and then subtracting the second result to the first.

What follows is the general definition.



(a) Determinant of the standard basis.



(b) The determinant of $A\hat{i}$ and $A\hat{j}$

Definition 2.3 (Determinant). Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$. The determinant of A is defined by

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij})$$

where M_{ij} is the matrix A where the row i and the column j have been removed.

We now list a few examples of linear transformations.

2.4 Linear Combinations and Linear Independence

We have just seen that in any vector space, vectors can be added together to combine their contribution into a new vector (Figure 2a). Moreover, vectors can be rescaled using the scalar multiplication (Figure 2b).

The process of obtaining a third vector from a *combination* of other two vectors, be that via addition and/or rescaling is called a *linear combination*. In other words, a vector v is a linear combination of a sequence of vectors if there exists a sequence of scalars with which we can rescale each vector such that when we sum these rescalings together we obtain the vector v .

Definition 2.4 (Linear Combination). Given a sequence of vectors v_{i_i} the vector v is a linear combination of $\{v_{i_i}\}_i$ if it exists a sequence s_{i_i} of scalars such that

$$v = \sum_i s_i v_i$$

For example, in Figure 2a the vector w can be written as a linear combination of the vectors $u, v \in V$. In this case, we say w is *linearly dependent* from v and u . In other terms, we can also say that $v + u - w = 0$.

Definition 2.5 (Linear Dependency). Given a sequence of vectors v_{i_i} a vector w is a linearly dependent from $\{v_{i_i}\}_i$ if it exists a sequence s_{i_i} of scalars such that w is a linear combination of v_{i_i} .

Two vectors are *linearly independent* if they are not linearly dependent. That is, if it is not possible to write one in terms of the other. Another way of formulating linear independence is by saying that two vectors are linearly independent if the only way to make one equal to the other is to rescale them to zero, that is the only scalar making them equal is zero.

Definition 2.6 (Linear Independence). A sequence of vectors $v_1, \dots, v_n$ from a vector space V is linearly independent, such that if

$$a_1 v_1 + a_2 v_2 + \dots + a_k v_k = 0$$

then $a_1 = a_2 = \dots = a_n = 0$.

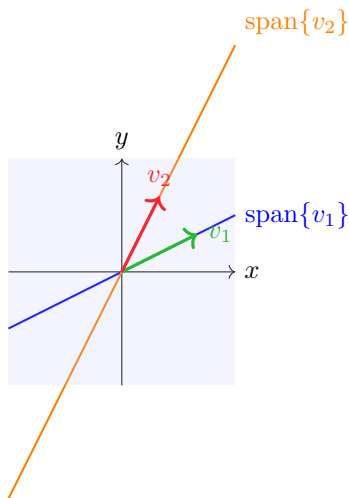


Figure 7: Span of $\{v_1, v_2\} = \mathbb{R}^2$

Essentially, in $\mathbb{R}^2$ if two vectors do not share the same direction they are linearly independent.

Now, given a set of vectors W , one may ask what is the set of vectors generated by these vectors by all possible linear combinations of these. This set, written $\text{span}(W)$, is called the *span* of W and it is simply the set of all possibly linear combinations of the vectors in W :

$$\text{span}(W) = \{a_1 w_1 + a_2 w_2 + \cdots + a_n w_n \mid w_1, \dots, w_n \in W, a_1, \dots, a_n \in S\}$$

For example, the span of the two vectors v_1 and v_2 as in Figure 7 is the entire space $\mathbb{R}^2$. We will see shortly that the reason why this happens is because the two vectors v_1 and v_2 are linearly independent and the size of the space is 2. In fact, any two linearly independent vectors in $\mathbb{R}^2$ would generate the space including the pair $\hat{i}$ and $\hat{j}$. Notice that the span of a single vector is the entire line along that vector (Figure 7).

This leads to the notion of subspace, that is essentially a subset of a vector space that is still a vector space.

Definition 2.7 (Vector Subspace). *Let V be a vector space over a field $\mathbb{F}$. A subset $W \subseteq V$ is a subspace of V if W is a vector space under the operations inherited from V .*

For example, given some vectors W in $\mathbb{R}^2$, the span of W is the subspace of $\mathbb{R}^2$ generated by the vectors in W . Of course the span of all the vectors in $\mathbb{R}^2$ also generate the whole space, but we are interested in the minimal set of vectors which generate the whole space from whence they came from. This leads to the notion of basis vectors. Intuitively, for a vector space V , a basis is the minimal set of vectors which generate V .

Definition 2.8 (Basis). *A basis for a vector space V is a sequence e_i of linearly independent vectors such that every $v \in V$ can be written as a linear combination of vectors e_i .*

In other words, the base E is the smallest subset of linearly independent vectors such that $\text{span}(E) = V$.

The *standard basis* for the Euclidean space $\mathbb{R}^2$ are the vectors $\hat{i}$ and $\hat{j}$. For an n -dimensional vector space over the reals, that is $\mathbb{R}^n$, the standard basis is the set of vectors $\{\hat{e}_i\}$ defined as

$$\hat{e}_{i,j} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{otherwise} \end{cases}$$

For example, the standard basis of the 4-dimensional real vector space $\mathbb{R}^4$ is

$$\mathcal{B} = \{e_1, e_2, e_3, e_4\},$$

where

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

By definition for every vector $v \in \mathbb{R}^n$ there exists a set of scalars such that v can be written as a linear combination of the basis vectors. Assume $v = (v_1, v_2, v_3)$ then clearly v can be written as a linear combination of the standard basis where the scalars are the components of v itself:

$$v = v_1 e_1 + v_2 e_2 + v_3 e_3.$$

Exercise 2.3. Show that the vector $v = (3, 5, 1)$ can be written as a linear combination of the standard basis.

The notion of basis vectors are central in linear algebra. For example, it is useful to prove that every linear transformation is essentially a matrix and viceversa.

Lemma 1. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation then there exist a matrix $A \in \mathbb{R}^{m \times n}$ such that

$$T(v) = Av \quad \text{for all } v \in \mathbb{R}^n.$$

for all $v \in \mathbb{R}^n$.

Proof. Recall, that every vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of these basis vectors:

$$v = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n.$$

Using linearity of T (Definition 2.2), we have

$$T(v) = T(v_1 e_1 + \cdots + v_n e_n) = v_1 T(e_1) + \cdots + v_n T(e_n).$$

Now we define the matrix A by taking the images of the standard basis vectors as its columns:

$$A = [T(e_1) \quad T(e_2) \quad \cdots \quad T(e_n)].$$

As an example, consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (2x + y, 3x - y).$$

and compute

$$T(1, 0) = (2, 3), \quad T(0, 1) = (1, -1).$$

Thus, the matrix representation of T is

$$A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}.$$

This concludes the proof. □

2.5 The norm of a vector

The *norm* (or length) of a vector $\vec{v}$ in a vector space V is simply given by Pythagoras' theorem:

$$\|\vec{v}\| = \sqrt{v_1^2 + \dots + v_n^2}$$

where $v_1, \dots, v_n$ are the components of the vector, or in other terms, the projections of the vector onto the respective axis. An example for $\mathbb{R}^2$ is depicted in Figure 10a.

The norm gives rise to the *distance* between two vectors $\vec{u}$ and $\vec{v}$ which can be calculated as the norm of the difference of the two vectors:

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$$

To see this consider how to compute the distance of two vectors u and v in the space $\mathbb{R}^2$ as in Figure 10b. The difference between the two vectors is a new vector calculated by subtracting component-wise, then the norm is the length of the difference of the two vectors hence it is the distance between the two points.

2.6 Invertible Transformations

A transformation A when applied to a vector x causes the vector x to move resulting in another vector, say b . An interesting question to ask is whether it is possible to reverse the effect of the matrix A , that is whether there exists a matrix, named A^{-1} , such that $A^{-1}b = x$. Notice that since linear transformations are functions on a vector space there has to be a unique x such that $Ax = b$ or it won't be possible to construct A^{-1} .

Geometrically, for a matrix $A = [a_1, a_2, \dots, a_n]$ the equation $Ax = b$ asks whether the vector b lies in the column space of A , that is $\text{col}(A)$:

$$\text{col}(A) = \text{span}(a_1, a_2, \dots, a_n)$$

which is all possible linear combinations of the column vectors of the matrix A . In other words, $\text{col}(A)$ is the set of all reachable outputs you can get from the transformation A , that is the *range* of the linear transformation.

Now consider the space $\mathbb{R}^2$ and consider two linearly independent vectors v and u , for example, as those in Figure 2a. If $A = [v, w]$ then the column space would generate the whole space $\mathbb{R}^2$ simply because for a 2 dimensional space, if we have two linearly independent vectors these form a basis. This means that any other vector b would be linearly dependent from the columns of A as b is contained in $\text{col}(A)$ and therefore can be written as a linear combination of the columns of A . Now consider the single column matrix $B = [v]$. The columns of this matrix do not generate the whole space, so there is no guarantee that if we chose a vector b there would be an x from whence b came. Finally, consider the matrix $C = [v, w, u]$ from Figure 2a. While $\text{col}(C)$ generates the whole space, it is not a basis of $\mathbb{R}^2$ because w is linearly dependent from the other two vectors. This means that a vector $b \in \mathbb{R}^2$ can be written both as a linear combination of v and u , but also as linear combination of v and w . This implies that when looking for a vector x such that $Cx = b$ there are at least two options: choosing the x which nullifies the column w or the column u (but there are many other options!).

This brings us to the notion of *rank*. The *rank* of a matrix is the maximum number of linearly independent rows or columns in the matrix. In other words, it measures how many rows or columns contain genuinely new information. The formal definition is as follows:

Definition 2.9 (Rank). Let $A \in \mathbb{R}^{m \times n}$. The rank of A , denoted $\text{rank}(A)$, is the dimension of the column space of A :

$$\text{rank}(A) = \dim(\text{Col}(A)).$$

where the column space of A , denoted $\text{Col}(A)$, is the subspace of $\mathbb{R}^m$ spanned by the columns of A .

Not that the dimension of the column space is exactly the number of linearly independent column vectors.

To better understand this idea, consider that each column of a matrix can be viewed as a vector. If one column can be written as a linear combination of other columns, then it does not contribute any new information. For example, consider the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

The second column is equal to twice the column, so there is only one linearly independent column. Therefore, the rank of this matrix is 1. This means that the 2×2 matrix transforms vectors in $\mathbb{R}^2$ into vectors in $\mathbb{R}$. Since there are many more vectors in $\mathbb{R}^2$ than in $\mathbb{R}$ then it is simply not possible to map every vector in $\mathbb{R}^2$ to just one vector in $\mathbb{R}$, hence A is not invertible.

Lemma 2. *Let a matrix $A \in \mathbb{R}^{n \times m}$ then A is invertible if and only if it has rank m .*

From the perspective of linear transformations, suppose a matrix represents a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The rank of the matrix is equal to the dimension of the image (or range) of T which is the subspace of $\mathbb{R}^m$ of vectors mapped by T . In geometric terms, it tells us how much of the output space is actually reached. A rank of 3 means the image is three-dimensional; a rank of 1 means that all outputs lie along a single line; and rank 0 means everything is mapped to the zero vector. Full rank indicates that the transformation does not collapse any dimensions beyond what is forced by the size of the matrix.

Equivalently, A is *singular* if $\det(A) = 0$ or $\text{rank}(A) < n$. If $\det(A) \neq 0$ (full rank), the matrix is called *nonsingular* or invertible. A singular matrix “collapses” space along one or more directions. Its column space does not fill $\mathbb{R}^n$, so the linear transformation A cannot be reversed. Geometrically, it might squash a plane into a line, or a volume into a plane, losing information along some dimensions.

For a square $n \times n$ matrix A , the rank also determines invertibility:

- If $\text{rank}(A) = n$, all columns are independent, the column space spans $\mathbb{R}^n$, and A is *invertible*.
- If $\text{rank}(A) < n$, some columns are dependent, the column space does not fill $\mathbb{R}^n$, and A is *singular* (non-invertible).

Example 2.1. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 3 & -2 \end{pmatrix}$. We have to find three coefficients such that

$$c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_3 \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

Hence we have to solve the system

$$\begin{cases} c_1 + 2c_2 + 3c_3 = 0 \\ -c_1 + 3c_2 - 2c_3 = 0 \end{cases}$$

which has solution $(c_1, c_2, c_3) = (13, 1, -5)$. Hence one of the columns is linearly dependent from the others. Say that $(2, 3)$ is linearly dependent from the other two vectors, that is

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = -13 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + 5 \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

Then we delete the column $(2, 3)$ and check if the rest is linearly independent. The second system is

$$\begin{cases} c_1 + 3c_2 = 0 \\ -c_1 - 2c_2 = 0 \end{cases}$$

From which we deduce that $c_1, c_2 = 0$, hence A has rank 2.

2.6.1 Diagonalisation and Changes of Basis

In linear algebra, a *change of basis* means describing the same vector using a different coordinate system. The vector itself does not change; only the way we represent it numerically changes. In $\mathbb{R}^2$, the standard basis is

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and every vector v can be written uniquely as a linear combination of these two vectors, for example $v = ae_1 + be_2$, where a, b are the coefficients of the vector v . Now consider the matrix

$$P = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$$

whose columns are $v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Since these two vectors are linearly independent, they form a new basis of $\mathbb{R}^2$ — albeit not an either orthogonal or normal one — hence, every vector v can be also be written as a linear combination of the basis vectors in P

$$v = xp_1 + yp_2$$

However, for the same vector v the coefficients x, y will not be the same as a, b which means that despite being the same vector, from the point of view of a different basis the vector has different coordinates. The equation above can be rewritten as

$$v = P \begin{pmatrix} x \\ y \end{pmatrix} \tag{3}$$

Then to find the coefficients of the vector v we have to compute the inverse P^{-1} and compute

$$\begin{pmatrix} x \\ y \end{pmatrix} = P^{-1}v$$

The vector obtained, namely $\begin{pmatrix} x \\ y \end{pmatrix}$, is the vector v as if v was given in the new basis P , and $\begin{pmatrix} x \\ y \end{pmatrix}$ is how the vector looks like in the standard basis. Similarly, to convert a vector $\begin{pmatrix} x \\ y \end{pmatrix}$ from the standard basis to the new basis we have to use the formula (3).

In this sense, the matrix P can be viewed as a *change of basis matrix*.

Now consider the rotation matrix R as in Figure 4a. This transformation operates in the standard basis interpretation of $\mathbb{R}^2$. We now want interpret a rotation as if the space was seen as from the lenses of a new basis whose vectors yield the linear transformation P . In other words, we want a 90° rotation matrix in the new basis.

Given a vector v in the new basis given by P , we have to transform it into a vector in the standard basis by multiplying by P . At this point we can rotate it using the rotation R in the standard basis and then we can convert the rotated vector into a vector in the new basis given by P . This translates to the following:

$$P^{-1}RPv$$

In particular, if P is defined as in the previous example we obtain the follow expression:

$$\overbrace{\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}^{-1}}^{\text{Change of basis}} \underbrace{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}_{\text{Rotation}} \overbrace{\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}}^{\text{Change of basis}} v$$

Note, that the composition of matrices $P^{-1}RP$ is the rotation matrix in the new basis we were looking for.

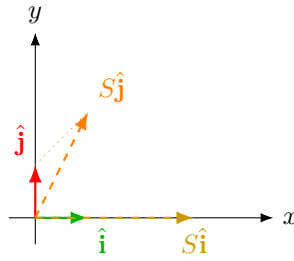


Figure 8: Eigenvectors and Eigenvalues

2.7 Eigen decomposition

Consider the standard basis vectors for $\mathbb{R}^2$ (this is just a subset of the standard basis in $\mathbb{R}^3$ as in Figure 2c) and consider again the shear matrix as in Figure 4c. After applying the shear to each vector $\hat{i}$ and $\hat{j}$ we can notice that certain vectors get knocked off their trajectory. Formally, for a vector v , the applying a transformation S to v means computing Sv which may or may not belong to the span of the original vector v , which is formed by all the possible rescalings of the vector $\hat{i}$.

In particular, for the standard basis $S\hat{i}$ belongs to the span of $\hat{i}$ and $S\hat{j}$ does not belong to the span of $\hat{j}$ as in Figure 8. This leads to the observation that there are certain vectors of the space which retain their trajectory and others that do not.

For a linear transformation, the vectors which retain their trajectory are called *eigenvectors* and for each *eigenvector* its rescaling factor is called the eigenvalue of the eigenvector.

Definition 2.10. Let A be an $n \times n$ matrix. A nonzero vector v is called an eigenvector of A if there exists a scalar λ such that

$$Av = \lambda v.$$

The scalar λ is called the eigenvalue corresponding to the eigenvector v .

This equation means that when the matrix A acts on the vector v , the result is simply a scalar multiple of v . In other words, the direction of v does not change under the transformation; it is only stretched or compressed (and possibly reversed if λ is negative). Most vectors change direction when multiplied by a matrix, but eigenvectors do not. To find eigenvalues, we rewrite the equation as

$$Av = \lambda v \implies (\lambda I - A)v = 0.$$

Notice that the problem of finding eigenvalues and eigenvectors is reduced to solve the equation $(\lambda I - A)v = 0$. Now, as we have seen in Section 2.6 this equation has a unique solution if the determinant is nonzero. However, if this was the case there would be only one v such that $(\lambda I - A)v = 0$, and that would be $v = 0$. Thus, for this equation to have a nontrivial solution $v \neq 0$, the matrix $A - \lambda I$ must be have a nonzero determinant. Therefore, we require

$$\det(\lambda I - A) = 0.$$

This equation is called the *characteristic equation* of A , and its solutions are the eigenvalues. Once the eigenvalues are found, the corresponding eigenvectors are non-zero vectors obtained by solving

$$(A - \lambda I)v = 0$$

for each eigenvalue λ .

Example 2.2. The task is to find the eigenvalues of the matrix

$$A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}.$$

The eigenvalues are the solutions to the equation $\det(\lambda I - A) = 0$, which we write as

$$\begin{vmatrix} \lambda - 3 & 0 \\ -8 & \lambda + 1 \end{vmatrix} = 0$$

from which we obtain

$$(\lambda - 3)(\lambda + 1) = 0.$$

This shows that the eigenvalues of A are $\lambda = 3$ and $\lambda = -1$. Thus, there are two eigenspaces of A , one for each eigenvalue. Let $x = (x_1, x_2)$ be an eigenvector of A corresponding to an eigenvalue λ iff x is a nontrivial solution of $\det(\lambda I - A)x = 0$, that is, of

$$\begin{pmatrix} \lambda - 3 & 0 \\ -8 & \lambda + 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $\lambda = 3$ then this equation becomes

$$\begin{pmatrix} 0 & 0 \\ -8 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

whose solution is $x_1 = 1/2t$ and $x_2 = t$. Thus the vector $(1/2, 1)$ is a basis for the eigenspace corresponding to $\lambda = 3$.

Exercise 2.4. Find the eigenbasis for the eigenvalue $\lambda = 1$

Exercise 2.5. Find the basis for the eigenspaces of the matrix

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

Geometrically, eigenvectors represent directions in which the linear transformation acts in a simple way—by pure scaling. Eigenvalues measure how much scaling occurs along those directions. These eigenvectors are the nonzero vectors in the null space of the matrix $\lambda I - A$. This null space is called the *eigenspace* of A corresponding to one particular λ .

Theorem 2.1 (Eigen decomposition). *if $A \in V^{n \times n}$ then the following statements are equivalent:*

- A is diagonalisable
- A has n linearly independent eigenvectors

Proof. Assume A has n linearly independent eigenvectors $p_1, p_2, \dots, p_n$ and that $\lambda_1, \lambda_2, \dots, \lambda_n$ are the corresponding eigenvalues. If we let $P = [p_1, p_2, \dots, p_n]$, and if we let D be the diagonal matrix that has $\lambda_1, \lambda_2, \dots, \lambda_n$ as its successive diagonal entries, then

$$AP = A[p_1, p_2, \dots, p_n] = [Ap_1, Ap_2, \dots, Ap_n] = [\lambda_1 p_1, \lambda_2 p_2, \dots, \lambda_n p_n] = PD.$$

Since the column vectors of P are linearly independent it follows that P is invertible by Lemma 2, so that this last equation can be rewritten as

$$P^{-1}AP = D$$

which is the decomposition of a diagonal matrix as wanted. □

The proof of Theorem 2.1 induces a decomposition procedure.

1. Find out if the matrix is diagonalisable by finding n linearly independent Eigenvectors. This means, find a basis for each Eigenspace and merge these basis vectors into a set S . If this set has fewer than n vectors, the matrix is not diagonalisable.
2. Form the matrix $P = (p_1 p_2 \dots p_n)$ where the vectors (p_i) are in S .
3. The matrix $P^{-1}AP$ will be diagonal and have eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ corresponding to the eigenvectors $p_1, p_2, \dots, p_n$ as its diagonal entries.

Exercise 2.6. Find (if they exists) the eigenvectors and eigenvalues of the rotation matrix, the shear and the projection matrix.

Theorem 2.2. If A is a $n \times n$ matrix, then the following statements are equivalent

- A is invertible.
- $Ax = b$ has exactly one solution for every $n \times 1$ matrix b .
- $Ax = 0$ has only the trivial solution.
- $\det(A) \neq 0$.
- The column vectors of A are linearly independent.
- The column vectors of A span $\mathbb{R}^n$.
- The column vectors of A are a basis of $\mathbb{R}^n$.
- A has rank n .
- A has nullity 0.
- The range of A is $\mathbb{R}^n$.
- $\lambda = 0$ is not an eigenvalue of A

In general, when the determinant $\det(\lambda I - A)$ is expanded, the result is a polynomial $p(\lambda)$ of degree n called the *characteristic polynomial* of A . For example, for the previous example

$$(\lambda - 3)(\lambda + 1) = \lambda^2 - 2\lambda - 3 = 0$$

which is a polynomial of degree 2. In general, the characteristic polynomial of a $n \times n$ matrix has the form

$$p(\lambda) = \lambda^n + c_1 \lambda^{n-1} + \dots + c_n$$

Since a polynomial of degree n has at most n distinct roots, it follows the equation

$$p(\lambda) = \lambda^n + c_1 \lambda^{n-1} + \dots + c_n = 0$$

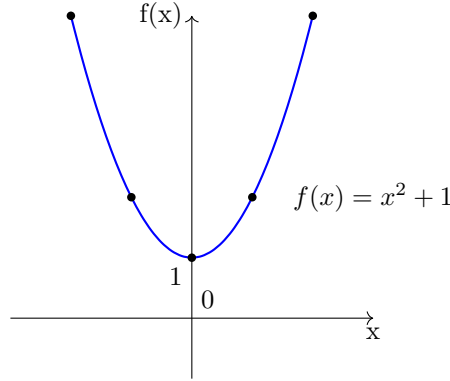
has at most n distinct solutions and, therefore, a matrix $n \times n$ has at most n distinct eigenvalues. However, the polynomial above does not necessarily have one solution in the real case. For example, the equation $\lambda^2 + 1 = 0$ has not solution in $\mathbb{R}^2$, but it has one in $\mathbb{C}^2$ by the fundamental theorem of algebra (2.4).

2.8 Complex Numbers

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as follows:

$$f(x) = x^2 + 1$$

Graphically, this function can be viewed as in the following picture



Obviously, if we ask the question, “Is there a value for x such that $f(x) = 0$ ” the answer would be negative: there is no x at which point the function f returns 0 because we would need to solve the equation $x^2 = -1$ and we know that there is no real number which multiplied by itself would yield a negative number. However, for some kinds of mathematical problems, such as the ones that we are going to treat in these notes, it would be useful to know that every function has a solution of this kind.

Complex numbers were invented to deal with this type of scenario by just assuming the square root of -1 exists. This imaginary number is called i . Now, if we take a look at the equation above again, we can compute $x^2 + 1 = 0$, which implies $x^2 = -1$, and thus $x = \sqrt{-1} = i$. Hence, i is our solution to the equation.

Algebraic Form Any complex number is formed by a real part and an imaginary part and is written as

$$z = a + bi \tag{4}$$

with $a, b \in \mathbb{R}$. Operations include addition, subtraction, multiplication, and division.

Polar Form A complex number $z = a + bi$ can also be represented in *polar form*, which expresses the number in terms of its *modulus* and *argument*.

In the complex plane, the complex number $z = a + bi$ is represented as a point (a, b) . The modulus r is the distance from the origin to this point and the argument θ is the angle formed with the positive real axis. The polar form provides a more convenient way to perform operations such as multiplication and division of complex numbers.

The diagram in Figure 9 illustrates a complex number $z = a + bi$ in the complex plane, where a is the real part ($\text{Re}(z)$), b is the imaginary part ($\text{Im}(z)$), $|z| = \sqrt{a^2 + b^2}$ is the magnitude (or modulus) of z and $\theta = \arg(z)$ is the argument (angle) of z . Clearly $a = r \cos \theta$ and $b = r \sin \theta$. So we can rewrite (4) as

$$z = r (\cos(\theta) + i \sin(\theta)) \tag{5}$$

Here’s some facts about complex numbers that are worth recalling.

Proposition 2.1. Assume $z = a + ib$ is a complex number, then:

- $\text{Re}(z)$ and $\text{Im}(z)$ are called real and imaginary parts of z , respectively,

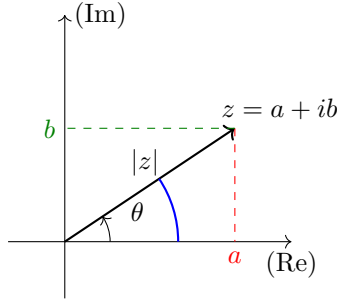


Figure 9: A geometric representation of complex numbers.

- $|z| = \sqrt{a^2 + b^2}$ is called the modulus or absolute value of z ,
- $z^* = a - ib$ is the complex conjugate of z ,
- $zz^* = |z|^2$,
- the angle θ in Figure 9 is the argument of z ,
- $\operatorname{Re}(z) = |z| \cos(\theta)$,
- $\operatorname{Im}(z) = |z| \sin(\theta)$,
- $z = |z|(\cos(\theta) + i \sin(\theta))$ is called the polar form of z .

Exponential Form The exponential form is based on the following theorem.

Theorem 2.3 (Euler's Formula). *For any real number θ , the following equation holds:*

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

where:

- e is Euler's number (the base of the natural logarithm),
- i is the imaginary unit ($i^2 = -1$),
- θ is a real number representing the angle (in radians),

Thus, the polar form

$$z = r(\cos(\theta) + i \sin(\theta))$$

can be conveniently written using Euler's formula as:

$$z = re^{i\theta}$$

where $e^{i\theta} = \cos(\theta) + i \sin(\theta)$ by Euler's formula.

As we mentioned above, when we work with complexes every polynomial function has a solution. This is made formal by the fundamental theorem of algebra and it is going to be a crucial result for describing models of quantum computation:

Theorem 2.4 (Fundamental Theorem of Algebra). *Every non-constant polynomial with complex coefficients has at least one complex root. In other words, for any polynomial $p(z)$ of degree $n \geq 1$, where*

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0,$$

with $a_n \neq 0$, there exists at least one complex number z_0 such that

$$p(z_0) = 0.$$

Example 2.3. *The characteristic equation of a general $n \times n$ matrix A defined as*

$$A = \begin{pmatrix} -2 & -1 \\ 5 & 2 \end{pmatrix}$$

is

$$\lambda^2 + 1 = 0$$

which has solution $\lambda = i$ and $\lambda = -i$.

2.9 Hilbert Spaces

2.9.1 The dot product

The *dot product* takes two vectors and returns a single scalar number, measuring how much they point in the same direction. More specifically, if w is a vector of length 1, the dot product of two vectors $v, w \in \mathbb{R}^n$, written $v \cdot w$, is the projection of the vector v onto the vector w . The dot product is thus related to the cosine of the angle θ between the two vectors. Specifically, for two vectors $u, v \in V$ as in Figure 10c, the dot product can be seen as the influence of the vector v onto the vector u . This depends of course on what is the angle θ between the two. Thus, the dot product $\langle \cdot, \cdot \rangle : V \times V \rightarrow V$ can be expressed by the following formula:

$$u \cdot v = \|u\| \cdot \text{proj}_u(v) = \|u\| \|v\| \cos(\theta)$$

where $\|u\|$ and $\|v\|$ are the magnitudes (norms) of the vectors, and θ is the angle between them.

As stated above, the dot product is thus useful to determine whether two vectors point in the same direction, in opposite direction or if they are orthogonal:

- $u \cdot v = 0$ then the vectors are *orthogonal*
- $u \cdot v < 0$ then the vectors are pointing in opposite direction
- $u \cdot v > 0$ then the vectors are pointing in the same direction

The definition given above implies knowledge of the angle between the two vectors. In linear algebra however is easier to work with coordinates. The dot product has an equivalent formulation in as the summation of the pointwise product of the coordinates.

Definition 2.11 (Dot Product). *For two vectors $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ in $\mathbb{R}^n$, the dot product is defined as:*

$$u \cdot v \triangleq \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

To show why this definition makes sense, assuming we are in $\mathbb{R}^2$, we compute as follows:

$$\begin{aligned}
u \cdot v &= \|u\| \|v\| \cos(\theta_w - \theta_v) \\
&= \|u\| \|v\| (\cos(\theta_w) \cos(\theta_v) + \sin(\theta_w) \sin(\theta_v)) \\
&= \|u\| \|v\| \cos(\theta_w) \cos(\theta_v) + \|u\| \|v\| \sin(\theta_w) \sin(\theta_v) \\
&= v_x w_x + v_y w_y
\end{aligned}$$

which shows that the dot product can be equivalently computed as the summation of the pointwise product of the coordinates.

Importantly, the dot product yields a metric, that is a notion of distance between two vectors. To see this compute the dot product between a vector $v \in \mathbb{R}^n$ and itself:

$$v \cdot v = v_1^2 + \dots + v_n^2$$

hence the norm of a vector v is just the square root of the dot product of the vector v with itself:

$$\|u\| = \sqrt{u \cdot u} = \sqrt{u_1^2 + \dots + u_n^2}$$

Now the distance between two vectors is simply the norm of the difference between the two:

$$d(\vec{u}, \vec{v}) = \|u - v\| = \sqrt{(u - v) \cdot (u - v)}.$$

This is depicted in Figure 10b. For example, for the vectors $\vec{u} = (2, 3, 4)$ and $\vec{v} = (1, 0, -1)$, to calculate their distance we first compute their difference component-wise:

$$\vec{u} - \vec{v} = (2 - 1, 3 - 0, 4 - (-1)) = (1, 3, 5)$$

then, we compute the norm of $\vec{u} - \vec{v}$:

$$\|\vec{u} - \vec{v}\| = \sqrt{1^2 + 3^2 + 5^2} = \sqrt{1 + 9 + 25} = \sqrt{35}$$

So, the distance between $\vec{u}$ and $\vec{v}$ is $\sqrt{35}$.

In general, the dot product has the following abstract definition.

Definition 2.12 (Dot product). *For a vector space V the dot product $(\cdot) : V \times V \rightarrow V$ is a binary operation such that the dot product has several important properties:*

1. $u \cdot v = v \cdot u$
2. $u \cdot (v + w) = u \cdot v + u \cdot w$
3. $(au) \cdot v = a(u \cdot v)$

Exercise 2.7. *Prove that the dot product in $\mathbb{R}^n$ satisfies Definition 2.12.*

2.9.2 Inner product

The inner product is a generalisation of the dot product that accounts for conjugate values. The inner product in the complex space $\mathbb{C}^n$ the Hermitian inner product:

$$\langle u, v \rangle = \sum_{k=1}^n u_k^* v_k$$

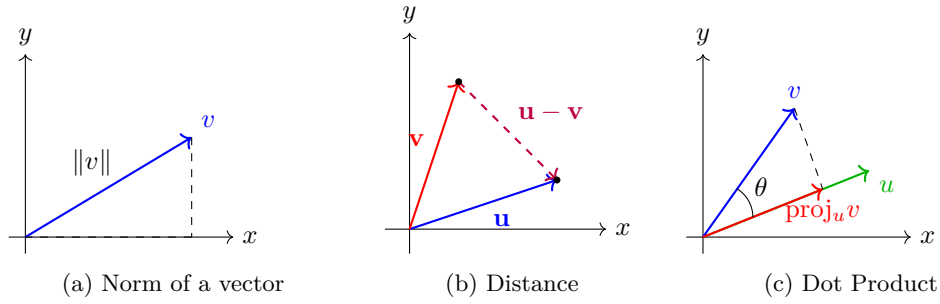


Figure 10: Norm and Dot Product

where u_k^* is the complex conjugate of the complex number u_k . This can be written more concisely by defining the conjugate transpose of a matrix. Notice that in the reals the conjugate of a real value is the value itself, so the inner product in $\mathbb{R}^n$ is just the dot product:

$$\langle u, v \rangle \triangleq u \cdot v$$

Now define the conjugate transpose of a square matrix as the matrix that is derived by first transposing the matrix itself and then conjugate all its complex entries, that is:

$$\overline{A} = (A^T)^*$$

Then the complex inner product of two vectors v, w of the same dimension can be defined simply obtained by the matrix-matrix multiplication between the conjugate transpose of v and w itself:

$$\langle v, w \rangle = \overline{v} w$$

Example 2.4. For $u = (1 + i, 2), v = (2, 1 - i)$ the inner product is calculated as

$$\langle u, v \rangle = (1 + i)^* \cdot 2 + 2^* \cdot (1 - i) = (1 - i) \cdot 2 + 2 \cdot (1 - i) = 4 - 4i.$$

So, the inner product of u and v is -2 .

The inner product is axiomatised as follows.

Definition 2.13 (Inner Product). Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. An inner product on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$$

(sometimes called a scalar product) satisfying the following properties for all $u, v, w \in V$ and all scalars $\alpha \in \mathbb{F}$:

1. $\langle v + w, z \rangle = \langle v, z \rangle + \langle w, z \rangle$
2. $\langle v, w + z \rangle = \langle v, w \rangle + \langle v, z \rangle$
3. $\langle v, \alpha w \rangle = \alpha \langle v, w \rangle$
4. $\langle \alpha v, w \rangle = \alpha^* \langle v, w \rangle$
5. $\langle v, w \rangle = \langle w, v \rangle^*$
6. $\langle v, v \rangle \geq 0$, with $\langle v, v \rangle = 0$ if and only if $v = 0$

Exercise 2.8. Prove the dot product is an inner product.

As it was the case for the dot product, the norm of a vector v can also be computed using the inner product. For the case of the reals this is straightforward since the inner product in $\mathbb{R}^n$ is the dot product. In the case of $\mathbb{C}^n$ we can just compute:

$$\begin{aligned}\|\vec{v}\| &= \sqrt{\langle v, v \rangle} \\ &= \sqrt{\bar{v}v} \\ &= \sqrt{v_1^* v_1 + \cdots + v_n^* v_n} \\ &= \sqrt{|v_1|^2 + \cdots + |v_n|^2}\end{aligned}$$

In particular, the norm of a vector in the complex case is the square root of the sum of the modulus of each entry. This relies on the property (2.1) of the complex numbers.

Thus, as it was the case for the dot product, the inner product induces a norm, and the norm induces a distance between vectors as before:

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$$

A property of the inner product that is going to be useful in quantum is the *Cauchy-Schwarz inequality* which states that in any inner product space,

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

for all vectors u and v . It says that the absolute value of the inner product can never exceed the product of the norms of the vectors. Geometrically, since

$$\langle u, v \rangle = \|u\| \|v\| \cos \theta,$$

the inequality simply reflects that $|\cos \theta| \leq 1$. Equality holds if and only if u and v are linearly dependent.

Finally, recall that the dot product was a useful mechanism to determine whether two vectors are orthogonal or normal, or both. The inner product inherits this idea from the real case and extends it to the complex case. This is useful to define the notion of an orthonormal basis.

Definition 2.14 (Orthogonal and Orthonormal Basis). *For a vectors space V with an inner product, an orthogonal basis is a family of elements $\{e_i\}$ with the following properties:*

- *they are pairwise orthogonal, i.e. $\langle e_i, e_j \rangle = 0$ for all $i \neq j$;*
- *every element $a \in H$ can be written as a linear combination of e_i*

Additionally, an orthogonal basis is orthonormal if all its vectors are of length 1, i.e. $\langle e_i, e_i \rangle = 1$, for all i .

This definition becomes even easier when we introduce the Kronecker δ function, pronounced the “Kronecker delta”.

Definition 2.15 (Kronecker δ). *The Kronecker delta is the function*

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

The Kronecker delta acts like an identity selector:

$$\sum_j \delta_{ij} a_j = a_i$$

It is the discrete analogue of the identity matrix entries. This is convenient to define orthonormal basis very concisely.

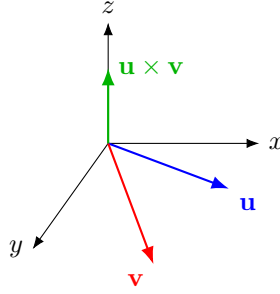


Figure 11: Cross product

Proposition 2.2. Let $\mathbb{B} = \{e_i\}$ be a set of basis vectors. Then $\mathbb{B}$ is orthonormal iff

$$\langle e_i, e_j \rangle = \delta_{ij}$$

for all i, j .

At this point we have built enough intuition on the inner product so that we are ready to define a Hilbert space.

Definition 2.16 (Hilbert Space). A Hilbert space is a vector space H equipped with an inner product such that the metric induced by the norm is complete, i.e. every Cauchy sequence converges.

For example, the Euclidean space $\mathbb{R}^n$ and the complex space $\mathbb{C}^n$ are Hilbert spaces as they are both equipped with an inner product and it can be proven that the metric is complete.

For the sake of these lecture notes, it is enough to understand a Hilbert space in terms of these two examples, however, the proper definition includes a wider class of vector spaces. In general, a Hilbert space is a mathematical structure that extends the familiar idea of Euclidean space to possibly infinite dimensions, while preserving key geometric notions like distance, angles, and orthogonality through an inner product. Importantly, the fact that a Hilbert space is “complete” means that limits of converging sequences remain within the space.

2.9.3 More Products

The *cross product* (or vector product) is a binary operation on two vectors in three-dimensional space $\mathbb{R}^3$ that produces another vector. For two vectors $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$, the cross product is defined as:

$$u \times v = \begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$$

where $\hat{i}, \hat{j}, \hat{k}$ are the standard basis vectors. Expanding the determinant:

$$u \times v = (u_2 v_3 - u_3 v_2) \hat{i} - (u_1 v_3 - u_3 v_1) \hat{j} + (u_1 v_2 - u_2 v_1) \hat{k}$$

The magnitude of the cross product is given by:

$$\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin(\theta)$$

where θ is the angle between the two vectors.

The *tensor product* is a generalization of the outer product (or dyadic product) and is used to construct higher-dimensional objects from two vectors or two matrices. For two vectors $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$, the tensor product $u \otimes v$ is an $m \times n$ matrix defined by:

$$\vec{u} \otimes \vec{v} = \begin{pmatrix} u_1 v_1 & u_1 v_2 & \cdots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \cdots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \cdots & u_m v_n \end{pmatrix}$$

Definition 2.17 (Kronecker Product). *Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$ two matrices. The Kronecker product of A and B is the block matrix*

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix} \in \mathbb{C}^{mp \times nq}.$$

Definition 2.18. *Let $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$ be column vectors. The outer product of u and v is the $m \times n$ matrix*

$$u \otimes v = uv^T = \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix} (v_1 \quad \cdots \quad v_n) = \begin{pmatrix} u_1 v_1 & \cdots & u_1 v_n \\ \vdots & \ddots & \vdots \\ u_m v_1 & \cdots & u_m v_n \end{pmatrix}.$$

Each column of uv^T is a scaled copy of u , and each row is a scaled copy of v^T .

2.10 Hermitian Matrices and the Spectral Decomposition Theorem

In this section we are interested in linear transformation that preserve the inner product of two vectors, but in a “complementary way”. For example, consider the standard basis of the Euclidean space $\mathbb{R}^2$ and consider the two basis vectors $\hat{i}$ and $\hat{j}$ as in Figure 12. The rotation matrix applied only to $\hat{i}$, as in Figure 12a moves the vector in the same position as $\hat{j}$. Hence the inner product $\langle R\hat{i}, \hat{j} \rangle = 1$. Their angle is 0° and both their lengths are 1 hence the contribution of $R\hat{i}$ onto $\hat{j}$ is 1. The matrix associated with R , written $R^\dagger$ and pronounced the Hermitian adjoint, is the unique matrix which transforms $\hat{j}$ such that it does not change the inner product with $\hat{i}$. In particular, in Figure 12b the Hermitian adjoint $R^\dagger \hat{j}$ in the same position as $\hat{i}$, thus maintaining the inner product.

We now define the Hermitian adjoint formally.

Definition 2.19 (Conjugate transpose / Hermitian transpose / Hermitian adjoint). *Let $A = (A_{ij}) \in \mathbb{C}^{n \times n}$ be a complex square matrix. The conjugate transpose of A , denoted $A^\dagger$, is the unique matrix which satisfies the equation*

$$\langle Av, w \rangle = \langle v, A^\dagger w \rangle \quad (6)$$

We are going to make extensive use of this fact in Section 2.11.

In the case of the complex vector space $\mathbb{C}^n$ the Hermitian adjoint is just the complex conjugate.

Proposition 2.3. *The Hermitian adjoint of a complex matrix is the conjugate transpose*

$$A^\dagger = (A^T)^*$$

Proof. Assume two vectors v, w , such that $\langle Av, w \rangle = \langle v, A^\dagger w \rangle$. By unfolding the definitions of inner product we get that $\overline{Av}w = \overline{v}A^\dagger w$, for all v, w . By the properties of the conjugate transpose $\overline{Av}w = \overline{v}\overline{A}w$. Since this is true for all w, v we can cancel them, thus obtaining $\overline{A} = A^\dagger$. $\square$

Corollary 1. *Let $A \in \mathbb{C}^{n \times m}$. The Hermitian adjoint is the matrix obtained by taking the transpose and then the complex conjugate of each entry:*

$$(A^\dagger)_{ij} = A_{ji}^*, \quad i = 1, \dots, n, \quad j = 1, \dots, m,$$

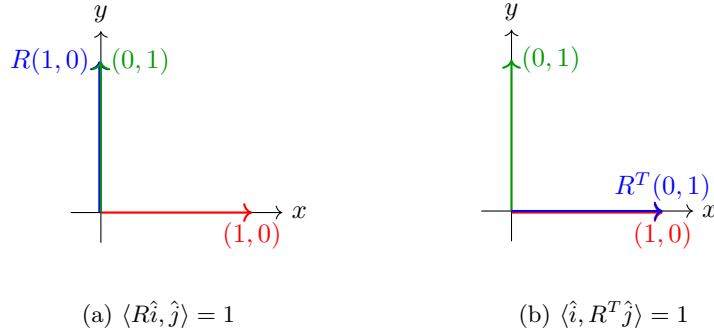


Figure 12: Hermitian adjoint

Exercise 2.9. Prove that the Hermitian adjoint satisfies the following properties:

1. $A^{\dagger\dagger} = A$
2. $(A + B)^{\dagger} = (A^{\dagger} + B^{\dagger})$
3. $(AB)^{\dagger} = B^{\dagger}A^{\dagger}$

Notice however, that even though $R^{\dagger}$ happens to coincide with R^{-1} , it is not in general true that the conjugate transpose is the inverse of the matrix. For example, the Hermitian adjoint of $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ is A itself since its entries are reals and the matrix is diagonal, but its inverse A^{-1} is $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$. This matrix however is particular since its Hermitian adjoint is itself, and it's called an Hermitian matrix.

Definition 2.20 (Hermitian Matrix). A square matrix $A \in \mathbb{C}^{n \times n}$ is called Hermitian if it is equal to its conjugate transpose:

$$A = A^{\dagger}$$

that is,

$$A_{ij} = A_{ji}^* \quad \text{for all } i, j = 1, \dots, n,$$

where A_{ji}^* denotes the complex conjugate of A_{ji} .

Hermitian matrices are real symmetric matrices. That is for a matrix A , all the entries A_{ij} are real and $A^T = A$. This is a very useful property to have as it admits a diagonal decomposition which we described in Section 2.6.1.

Theorem 2.5 (Spectral Decomposition Theorem). Let $A \in \mathbb{R}^{n \times n}$ be a real symmetric matrix (or $A \in \mathbb{C}^{n \times n}$ be Hermitian, i.e., $A = A^{\dagger}$). Then there exist orthonormal eigenvectors $v_1, \dots, v_n$ and real eigenvalues $\lambda_1, \dots, \lambda_n$ such that

$$A = \sum_{i=1}^n \lambda_i v_i v_i^T.$$

Equivalently, A can be written as

$$A = Q\Lambda Q^T,$$

where Q is an orthogonal (unitary) matrix whose columns are the eigenvectors of A , and Λ is a diagonal matrix whose diagonal entries are the eigenvalues of A .

Proof. To see this we explain how to construct Q and Λ . Since A is symmetric, its characteristic polynomial has at least one real root. Hence, there exists $\lambda_1 \in \mathbb{R}$ and a nonzero vector v_1 such that

$$Av_1 = \lambda_1 v_1.$$

Normalize v_1 so that $\|v_1\| = 1$. Let

$$V_1 = \{x \in \mathbb{R}^n : v_1^T x = 0\}$$

be the orthogonal complement of $\text{span}(v_1)$. We claim V_1 is closed under application of A . Indeed, for any $x \in V_1$,

$$v_1^T(Ax) = (Av_1)^T x = \lambda_1 v_1^T x = 0,$$

so $Ax \in V_1$.

Restrict A to V_1 . This restriction is again symmetric, so we can repeat the argument: there exists a unit eigenvector $v_2 \in V_1$ with eigenvalue λ_2 .

Continuing inductively, we obtain an orthonormal set $\{v_1, \dots, v_n\}$ of eigenvectors with corresponding real eigenvalues $\lambda_1, \dots, \lambda_n$.

We now construct Q as the matrix formed by the vectors v_i

$$Q = [v_1 \ \cdots \ v_n], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Then Q is orthogonal ($Q^T Q = I$), and

$$AQ = Q\Lambda.$$

Multiplying on the right by Q^T gives

$$A = Q\Lambda Q^T.$$

□

2.11 Unitary Matrices

A *unitary* matrix represents a transformation that preserves the geometry of the space. In particular, it does not change lengths or angles between vectors. Following this the matrix U is unitary and $x, y \in \mathbb{C}^n$, then

$$\|Ux\| = \|x\| \quad \text{and} \quad \langle Ux, Uy \rangle = \langle x, y \rangle.$$

The first equation states that the length of the vector does not change after applying the unitary transformation, while the second states that apply U to two vectors preserves their inner product. Thus, a unitary matrix can be thought of as a *rigid motion* in complex vector space: it may rotate vectors, reflect them, or change their complex phase, but it never stretches or distorts them.

For example, we consider the rotation matrix R and the matrix $N = \begin{pmatrix} 1 & 2 \\ 1.5 & 2 \end{pmatrix}$. The rotation matrix is a classic example of a unitary matrix whilst N is non-unitary. To see this it suffices to see what the respective matrices do on the standard basis vectors $\hat{i}$ and $\hat{j}$. The rotation sends the standard basis vectors to $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$, respectively, as in Figure 13a. On the other hand, the matrix N maps the standard basis vectors to two vectors with a smaller angle between them and different lengths as in Figure 13b.

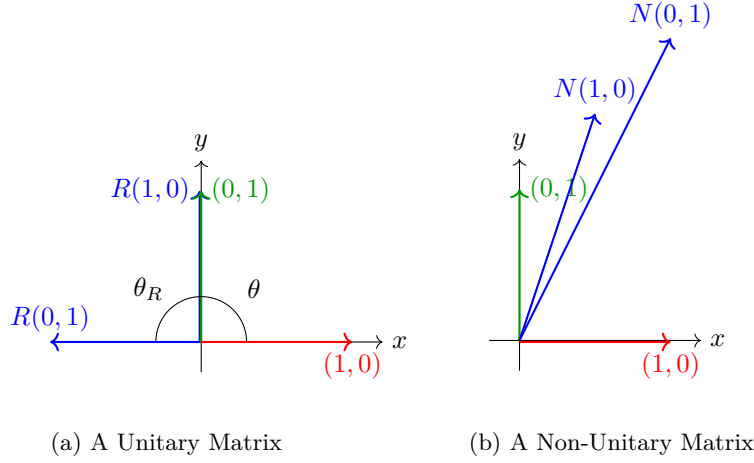


Figure 13: Unitary and Non-Unitary Matrices

From these intuition we can derive the formal definition of a unitary matrix. First recall that the inner product in a complex space is defined as $\langle x, y \rangle = x^\dagger y$. Then we compute as follows:

$$\begin{aligned}
 \langle Ux, Uy \rangle &= \langle x, y \rangle \\
 \iff (Ux)^\dagger Uy &= x^\dagger y \\
 \{ \text{By Exercise 2.9} \} \\
 \iff x^\dagger U^\dagger Uy &= x^\dagger y \\
 \{ \text{Bring everything to one side} \} \\
 \iff x^\dagger U^\dagger (Uy) - x^\dagger y &= 0 \\
 \{ \text{factor out } x^\dagger \} \\
 \iff x^\dagger (U^\dagger (Uy) - y) &= 0 \\
 \{ \text{factor out } y \} \\
 \iff x^\dagger (U^\dagger U - I)y &= 0
 \end{aligned}$$

Now if we let $A := U^\dagger U - I$, this amounts to solve the equation $x^\dagger Ay = 0$, for all vectors x, y . Now, since this has to hold for all vectors x, y it has to hold also for the standard basis vectors $\{e_i\}$. Since $e_j^\dagger A e_i = A_{i,j}$ then, for all i, j , $A_{i,j} = 0$ which implies that $A = 0$. Hence we get

$$U^\dagger U - I = 0$$

This leads to the formal definition of the a unitary matrix.

Definition 2.21 (Unitary Matrix). *A matrix $U \in \mathbb{C}^{n \times n}$ is called unitary if*

$$U^\dagger U = U U^\dagger = I,$$

where $U^\dagger$ denotes the conjugate transpose of U , and I is the identity matrix.

It follows that the determinant of a unitary matrix is always 1 and its eigenvalues are also always 1.

Furthermore, a matrix U is unitary if and only if its columns (and rows) form an orthonormal basis of $\mathbb{C}^n$, i.e.

$$\langle u_i, u_j \rangle = \delta_{ij}.$$

Property	Hermitian	Unitary	Diagonal
Definition	$A = A^\dagger$	$A^\dagger A = I$	Diagonal entries $\neq 0$
Geometric mean.	Scaling on orthogonal dir.	Rotation/reflection	Independ. scaling on axes
Eigenvalues	Real	$ \lambda = 1$	Diagonal entries
Eigenvectors	Orthonormal basis	Orthonormal basis	Standard basis
Diagonalizable	Yes (unitarily)	Yes (unitarily)	Already diagonal
Preserves length	No	Yes	No (unless ± 1)
Orthogonal eigenvec.	Yes	Yes	Yes
Inverse	If no zero eigenvalues	$A^{-1} = A^\dagger$	Invert diagonal entries
Spectrum	Real axis	Unit circle	Arbitrary

Table 1: Comparison of Hermitian, Unitary and Diagonal matrices

where δ_{ij} is the Kronecker δ .

2.12 Normal Matrices

Normal matrices generalise several kinds of well-behaved matrices including all those in Table 1, that is Hermitian, unitary and diagonal matrices. The formal definition is as follows:

Definition 2.22 (Normal Matrix). *A matrix is normal if it commutes with the conjugate transpose, that is*

$$\bar{A}A = A\bar{A}$$

Exercise 2.10. *Prove that Hermitian, unitary and diagonal matrices are normal.*

2.13 Trace of a Matrix

In a matrix, the diagonal elements measure how much each basis vector is stretched along itself. More precisely, if we consider a matrix A and a standard basis e_i then the entry A_{ii} specifies how much of the basis vector e_i is stretched along its direction. The trace of a matrix, essentially, sums all these contributions.

The *trace* of a square matrix $A \in \mathbb{C}^{n \times n}$ is defined as the sum of its diagonal elements:

$$\text{trace}(A) = \sum_{i=1}^n A_{ii}$$

Intuitively, the trace measures the total amount by which a linear transformation stretches space along its principal directions. Recall from Section 2.7 that each eigenvalue represents scaling the associated eigenvector along one direction. Essentially, the trace adds up all these scalings. For example, the trace of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

can be computed as $\text{trace}(A) = 1 + 5 + 9 = 15$. We list the properties of the trace operation:

- *Linearity:*

$$\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B) \quad \text{trace}(cA) = c \text{trace}(A)$$

- *Cyclic property:*

$$\text{trace}(AB) = \text{trace}(BA)$$

- *Invariance under similarity:*

$$\text{trace}(P^{-1}AP) = \text{trace}(A)$$

More in general, the cyclic property implies

$$\text{trace}(ABC) = \text{trace}(BCA) = \text{trace}(CAB)$$

Another useful result is that the trace equals the sum of the eigenvalues of A :

$$\text{trace}(A) = \sum_i \lambda_i$$

which holds even if A is not diagonal.

2.14 Dirac Notation

We conclude with introducing a notation for vectors and inner products which is summarized in Table 2. This is usually introduced with quantum mechanics albeit it is not strictly connected to it, in the sense, that we could have introduced at the beginning of these notes but we chose not to .

In Dirac notation, vectors in a complex Hilbert space are written as *kets* and *bras*. For a vector $\psi = \begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_n \end{pmatrix}$, a ket is simply written $|\psi\rangle$. Its conjugate transpose $\psi^\dagger$ is written as

$$\langle\psi| = (v_0^* \quad v_1^* \quad \cdots \quad v_n^*)$$

and pronounced “bra”. The reason why this notation is convenient is because the inner product arises as the product of a ket and bra, in words, a bra-ket (bracket):

$$\langle\phi|\psi\rangle = \langle\phi, \psi\rangle.$$

This is simply the matrix-vector multiplication of the row vector $\langle\psi|$ with the column vector $|\psi\rangle$. A linear transformation (or operator) A acting on a ket is written $A|\psi\rangle$. Moreover, applying a linear transformation on a $|\psi\rangle$ and then performing the inner product between $\langle\psi|$ and $A|\psi\rangle$ is written as follows:

$$\langle\phi|A|\psi\rangle = \langle\phi, A\psi\rangle.$$

Finally, the outer product of $|\phi\rangle$ and $\langle\psi|$ is

$$|\phi\rangle\langle\psi|$$

which is simply the matrix-matrix multiplication of a 1-column matrix $|\phi\rangle$ with a 1-row matrix $\langle\psi|$.

3 Quantum Mechanics

The first postulate states that Hilbert spaces are correct mathematical objects in which to interpret quantum state spaces.

Postulate 1. *Associated to any isolated physical system is a Hilbert space known as state space of the system. The system is completely described by its state vector which is a unit vector in the system’s state space.*

Notation	Description
z^*	Complex conjugate of the complex number z . Example: $(1 + i)^* = 1 - i$
$ \psi\rangle$	Vector, also known as a ket
$\langle\psi $	Vector dual to $ \psi\rangle$, also known as a bra
$\langle\phi \psi\rangle$	Inner product between the vectors $ \phi\rangle$ and $ \psi\rangle$
$ \phi\rangle \otimes \psi\rangle$	Tensor product of $ \phi\rangle$ and $ \psi\rangle$
$ \phi\rangle \psi\rangle$	Abbreviated notation for the tensor product
A^*	Complex conjugate of the matrix A
A^T	Transpose of the matrix A
$A^\dagger$	Hermitian conjugate (adjoint) of A
$\langle\phi A \psi\rangle$	Inner product between $ \phi\rangle$ and $A \psi\rangle$; equivalently between $A^\dagger \phi\rangle$ and $ \psi\rangle$
$\bar{A}$	The conjugate transpose $(A^T)^*$

Table 2: Dirac Notation from Nielsen and Chuang [3]

3.1 Quantum States and Quantum Superposition

A quantum state of a system is represented by a column vector whose indices are placed in correspondence with the classical states of that system. The entries are complex numbers such that the sum of the absolute values squared of the entries must equal 1. In other words, a quantum state is simply a column vector over the complexes such that its norm is 1:

$$\|v\| = \sqrt{\sum_{k=1}^n v_k^2} = 1$$

The qubit $|0\rangle$ is the column vector representing the *ground state* which is 0 with probability 1 and 1 with probability 0. The qubit $|1\rangle$ is the column vector representing the *excited state* which is 1 with probability 1 and 0 with probability 0.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

These two states form the *basis* of the complex Euclidean space $\mathbb{C}^2$ which is called the *computational basis*. This means that every other state can be written as a linear combination of these two states.

The linear combinations and computational basis are a fundamental concepts that allows us to define a *superposition* of states. Superposition describes the ability of a quantum system to exist simultaneously in multiple states. Unlike classical bits, which can only be in one state at a time (0 or 1), a qubit can be in a combination of both states at once. A quantum state in superposition is modelled by a linear combination as follows:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{7}$$

where α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. This is equivalent to saying that $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$. In other words, the qubit does not have a definite state but rather a probabilistic combination of states. Intuitively, when measured, the qubit collapses into one of the basis states, with probabilities determined by $|\alpha|^2$ and $|\beta|^2$.

Two examples of quantum states in superposition $|+\rangle$ and $|-\rangle$. These two states are interesting because they are both *equal superpositions* of $|0\rangle$ and $|1\rangle$, in the sense that the linear combinations have different coefficients, but they have a different *phase*:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

This means that the minus sign makes them behave differently, but one measures them they will look identical. Intuitively, $|0\rangle$ has half probability of being measured at $|0\rangle$ and half to be $|1\rangle$ and so is $|1\rangle$.

Exercise 3.1. Show that $|+\rangle$ and $|-\rangle$ form an orthonormal basis.

There are two things we can do with a quantum state: measure it or let it evolve unitarily without measuring it. We will explain evolution first.

3.2 Quantum Evolution and Quantum Gates

We now turn our attention to how quantum states can evolve. This first and perhaps most straightforward requirement is that, whatever the evolution of a particular quantum state may be, this has to preserve the sum of the probabilities for each outcome. In other words, if U is a linear operator acting on quantum system $|\psi\rangle$ then, if $\| |\psi\rangle \| = 1$ then also $\| U |\psi\rangle \| = 1$. By unfolding the definition of norm of a vector we obtain that the inner product $\langle U |\psi\rangle, U |\psi\rangle \rangle$ must be equal to $\langle \psi, \psi \rangle$, that is 1 (see Section 2.5). By Definition 2.19 the inner product $\langle U |\psi\rangle, U |\psi\rangle \rangle$ is equal to $\langle |\psi\rangle, U^\dagger U |\psi\rangle \rangle$ using its hermitian adjoint. We obtain the follow equation

$$\langle \psi | U^\dagger U | \psi \rangle = 1$$

for all $|\psi\rangle$. Since this has to hold for all $|\psi\rangle$ it has to hold in particular for the computational basis $\{|e_i\rangle\}$ which leads to the fact that $U^\dagger U$ must be the identity matrix.

In simple words, since quantum gates (or quantum evolutions) must be preserve probabilities, the operator U must be unitary as described in Section 2.11.

Postulate 2. The evolution of a closed quantum system is described by a unitary transformation. That is the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system t_2 by a unitary operator U which depends only on the times t_1 and t_2 .

Given a sequence of unitary operators indexed by a timestamp t , that is $U(t)$, and an initial state $|\psi(0)\rangle$, the time evolution of the system is given by:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

which is a sequence of vectors indexed by the timestamp.

Definition 3.1 (Quantum Gate). A quantum gate is a unitary operator acting on one or more qubits.

We list formally the properties arising from this definition:

- *Unitary:* $U^\dagger U = I$
- *Linear:* $U(a|\psi\rangle + b|\phi\rangle) = aU|\psi\rangle + bU|\phi\rangle$
- *Reversible:* $U^{-1} = U^\dagger$
- *Probability Conservation:* $\|U|\psi\rangle\| = \| |\psi\rangle \|$

3.2.1 Basic Quantum Gates

The *Pauli matrices* are a set of three fundamental 2×2 complex matrices that act on a single qubit. They are denoted by $\sigma_x, \sigma_y, \sigma_z$.

The first quantum gate we are going to see is the σ_x or bit-flip operator. The task is to come up with a matrix which maps $|0\rangle \mapsto |1\rangle$ and $|1\rangle \mapsto |0\rangle$. As we have seen in Section 2.2, we can just

construct the matrix by placing $|1\rangle$ in the first column and $|0\rangle$ in the second column, that is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Similarly, we can construct the bit-flip with a phase and the phase-flip operators:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In circuit notation, these operators are represented as single-qubit gates:

$$|\psi\rangle \text{---} \boxed{X} \text{---} \quad |\psi\rangle \text{---} \boxed{Y} \text{---} \quad |\psi\rangle \text{---} \boxed{Z} \text{---}$$

The Pauli matrices essentially correspond to rotations on the Bloch sphere, i.e. they generate rotations around the x , y , and z axes.

Exercise 3.2. Check that all these operators are Hermitian, unitary, their traces is 0 and their determinant is -1 . That is, for all $i \in \{x, y, z\}$,

- Hermitian: $\sigma_i^\dagger = \sigma_i$
- Unitary: $\sigma_i^\dagger \sigma_i = I$
- Self-inverse: $\sigma_i^2 = I$
- Trace and determinant: $\text{trace}(\sigma_i) = 0$, $\det(\sigma_i) = -1$

The *Hadamard gate*, denoted H , is a fundamental single-qubit quantum gate that creates superposition:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

In circuit diagrams, it is represented as:

$$|\psi\rangle \text{---} \boxed{H} \text{---}$$

To see why the Hadamard gate creates superposition, it is sufficient to apply it to the computational basis and observe that $H|0\rangle$ yields the superposition state $|+\rangle$ and, conversely, $H|1\rangle$ yields $|-\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle, \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle.$$

As all the other gates, the Hadamard also acts as a rotation on the Bloch sphere. Conversely, because the Hadamard is reversible, we also have

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle.$$

Exercise 3.3. Check that the Hadamard gate also satisfies the properties in Exercise 3.2.

Exercise 3.4. What are the eigenvalues and eigenvectors of H ?

3.2.2 Phase Operations

Phase operations act by modifying the complex phases of the amplitudes without changing their magnitudes. Consider a general qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. A *global phase* transformation multiplies the entire state by a common complex factor:

$$|\psi\rangle \longrightarrow e^{i\phi} |\psi\rangle. \tag{8}$$

Such a transformation has no physical or observable consequence, since all measurement probabilities depend only on $|\alpha|^2$ and $|\beta|^2$. For example,

$$H|0\rangle \longrightarrow e^{i\pi/3}H|0\rangle \quad (9)$$

represents the same physical state. In contrast, a *relative phase* transformation alters the phase between components of the superposition:

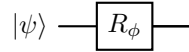
$$|\psi\rangle \longrightarrow \alpha|0\rangle + e^{i\theta}\beta|1\rangle. \quad (10)$$

This relative phase is physically meaningful and can affect interference phenomena, making it observable in subsequent measurements.

All standard single-qubit phase gates can be understood as special cases of a general phase rotation acting only on the $|1\rangle$ component:

$$R_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}, \quad R_\phi|\psi\rangle = \alpha|0\rangle + e^{i\phi}\beta|1\rangle. \quad (11)$$

In circuit form, this general phase rotation is represented as a single-qubit gate:



All the other phase operations are instances of this general transformation. In particular, the *Z gate* corresponds to a phase flip with $\phi = \pi$, the *Phase (S) gate* corresponds to $\phi = \frac{\pi}{2}$, and the *T gate* corresponds to $\phi = \frac{\pi}{4}$. Their matrix representations are:

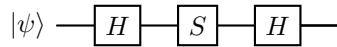
$$Z = R_\pi = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad S = R_{\pi/2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = R_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

These gates are represented in circuit diagrams as follows:



Therefore, all these gates implement the same fundamental operation: they leave the $|0\rangle$ component unchanged while applying a controlled phase shift to the $|1\rangle$ component. This highlights that phase gates do not change measurement probabilities directly, but instead modify interference patterns, which is essential for many quantum algorithms.

Quantum gates can be composed to construct new quantum gates, or derive old ones. In particular, the composition of quantum gates *HSH* produces a linear transformation whose square equals the bit-flip operation. This composition can be visualized as the sequential application of three gates:



Exercise 3.5. Prove that $(HSH)^2 = \sigma_x$.

3.3 Measurements

A general measurement is described by a set of operators $\{M_m\}$, where each operator corresponds to a possible outcome m . The central requirement is that if we could apply all of these operators obtaining the probability of each outcome the sum of the probabilities for each outcome needs to be equal to 1.

Assume we have a collection of operators $\{M_m\}$ and we apply one of these M_m to a state $|\psi\rangle$:

$$|\phi_m\rangle = M_m|\psi\rangle. \quad (12)$$

The probability of $|\phi_m\rangle$ is given by the norm $\|M_m |\psi\rangle\|$. As we did in Section 3.2, the norm is computed using the inner product $\langle\psi|M_m^\dagger M_m|\psi\rangle$. The requirement is that the sum of all of these inner products must be 1, formally:

$$\sum_m \langle\psi|M_m^\dagger M_m|\psi\rangle = 1,$$

for all $|\psi\rangle$. Since the inner product preserves addition (2) and by factoring out ψ the former is equivalent to $\langle\psi|(\sum_m M_m^\dagger M_m)|\psi\rangle = 1$. Now since this has to be true for all $|\psi\rangle$, it has in particular, to be true for the computational basis $\{|i\rangle\}$, hence:

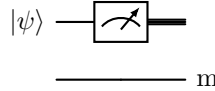
$$\sum_m M_m^\dagger M_m = I \quad (13)$$

ensuring that the total probability over all outcomes is 1.

If outcome m occurs, the measurement acts linearly on the state: $M_m |\psi\rangle$. In general, the resulting state is not normalized. The norm of the state is: $\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}$. To obtain a valid quantum state, we must normalize the vector. Therefore, the post-measurement state becomes:

$$|\psi_m\rangle = \frac{M_m |\psi\rangle}{\|M_m |\psi\rangle\|} = \frac{M_m |\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}. \quad (14)$$

In circuit terms, one can think of the measurement as producing a classical outcome while collapsing the quantum state:



Intuitively, the operator M_m extracts the component of the state corresponding to outcome m . The norm of this component gives the probability of that outcome. After the measurement, the state is renormalized to reflect the fact that outcome m has occurred.

These observations lead to the third postulate of quantum mechanics:

Postulate 3. *Quantum measurements are described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is given by*

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle.$$

The state after the measurement is

$$\frac{M_m |\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}.$$

3.3.1 Measurement in the Computational Basis

The simplest kind of measurement is called measurement in the computational basis and it is obtained by using the measurement operators $M_i = |i\rangle\langle i|$. For example, for a state $|\psi\rangle = a|0\rangle + b|1\rangle$ we have the measurement operators $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$. By using Postulate 3, the probability of the outcome 0 is obtained by computing $p(0)$:

$$p(0) = \langle\psi|M_0^\dagger M_0|\psi\rangle = \langle\psi|\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}|\psi\rangle = \langle\psi|\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}|\psi\rangle = a^*a = a^2$$

and the probability of the outcome 1 is obtained by computing $p(1)$ which is b^2 .

After the measurement the state collapses to one of the two computational basis states.

Definition 3.2 (Measurement in the Computational Basis). *Given a quantum state $|\psi\rangle$ in the computational basis $|e_i\rangle$, for a sequence $i \in \{1, \dots, n\}$ defined as*

$$|\psi\rangle = \alpha_1 |e_1\rangle + \dots + \alpha_n |e_n\rangle$$

The measurement in the computational basis states that upon measurement the state $|\psi\rangle$ collapses to the state $|e_i\rangle$ with probability $\|\alpha_i\|^2$.

It follows directly that the sum of the probabilities for each outcome is $\sum_{j=0}^{n-1} \|\alpha_j\|^2 = 1$.

Measurement in quantum mechanics is the process of observing a quantum system, causing it to collapse into one of its basis states. For a qubit in the state $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, measurement in the computational basis collapses the state $|\psi\rangle$ to either $|0\rangle$ with probability $|\alpha|^2$ or $|1\rangle$ with probability $|\beta|^2$.

For example, measuring the state $|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$ has equal probabilities of yielding $|0\rangle$ or $|1\rangle$, that is 50%. Similarly, measuring the state $|\psi\rangle = |0\rangle$ always yields $|0\rangle$ with 100% probability.

For an n -qubit system, the computational basis states are $|b_1 b_2 \dots b_n\rangle$, where $b_i \in \{0, 1\}$. A measurement collapses the system to one of these 2^n states. Consider the two-qubit state:

$$|\psi\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle.$$

If measured in the computational basis the system collapses to either $|00\rangle$ with probability of $|\frac{1}{\sqrt{2}}|^2 = 0.5$ or $|11\rangle$ with probability $|\frac{1}{\sqrt{2}}|^2 = 0.5$.

3.3.2 Projective Measurements

Definition 3.3 (Projective Measurement). *A projective measurement with m possible outcomes, is a collection of projectors $P_1, \dots, P_m$, pairwise orthogonal, that is $P_i P_j = 0$ if $i \neq j$, and, moreover, that sum identity $\sum_{j=1}^m P_j = I$.*

3.3.3 Distinguishing Quantum States

Distinguishing quantum states is a fundamental (and surprisingly subtle) problem in quantum mechanics. Suppose a quantum system is prepared in one of several known states:

$$\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\} \tag{15}$$

The goal is to determine which state was prepared by performing a measurement. A quantum system is prepared in one of several known states $\{|\psi_1\rangle, |\psi_2\rangle, \dots\}$, but the observer does not know which one. The goal is to determine the prepared state by performing a measurement.

The task is to design the measurement so that the outcome m correctly identifies the state $|\psi_i\rangle$ with high probability.

Perfect discrimination is possible if and only if the states are orthogonal. If the states are non-orthogonal, errors are unavoidable, and one must either minimize the probability of error, or allow for inconclusive outcomes. Thus, distinguishing quantum states is a problem of statistical inference: measurement outcomes provide partial information that is used to infer the unknown prepared state.

Assume we have two orthogonal states:

$$\langle \psi_1 | \psi_2 \rangle = 0 \tag{16}$$

and define measurement operators:

$$M_i = |\psi_i\rangle \langle \psi_i| \tag{17}$$

for $i \in \{1, 2\}$ which satisfying the properties

$$p(i|\psi_i) = 1 \quad (18)$$

$$p(j|\psi_i) = 0 \quad (j \neq i) \quad (19)$$

Assume we have two states $|\psi_1\rangle$ and $|\psi_2\rangle$ such that are not orthogonal, that is

$$\langle \psi_1 | \psi_2 \rangle \neq 0, \quad (20)$$

then perfect discrimination is impossible. The proof is given by contradiction. Suppose we attempt to distinguish $|\psi_1\rangle$ and $|\psi_2\rangle$ perfectly. Then there must exist measurement operators M_1, M_2 such that:

$$\langle \psi_1 | M_2^\dagger M_2 | \psi_1 \rangle = 0 \quad (21)$$

$$\langle \psi_2 | M_1^\dagger M_1 | \psi_2 \rangle = 0 \quad (22)$$

which implies:

$$M_2 |\psi_1\rangle = 0, \quad M_1 |\psi_2\rangle = 0 \quad (23)$$

By linearity, this leads to a contradiction unless the states are orthogonal.

If perfect discrimination is not possible, one can minimize the probability of error. For two states with prior probabilities p_1, p_2 , the optimal error is given by the Helstrom bound:

$$P_{\text{error}} = \frac{1}{2} \left(1 - \sqrt{1 - 4p_1 p_2 |\langle \psi_1 | \psi_2 \rangle|^2} \right) \quad (24)$$

3.4 Multiple Quantum Systems

Postulate 4. *The state space of a composite physical system is the tensor product of the state spaces of the component physical system. Moreover, if we have systems numbered $1, 2, \dots, n$, and a system I is prepared in the state $|\psi_i\rangle$ then the joint state of the total system is*

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle$$

In quantum mechanics, when two quantum systems are combined, their states are described in a *tensor product* space. Suppose you have two quantum systems, A and B , with state spaces $\mathcal{H}_A$ and $\mathcal{H}_B$. The state of the combined system is represented in the tensor product space:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$$

The tensor product is a way of combining the state spaces of two systems into a larger state space. If $|\psi_A\rangle$ is a state in $\mathcal{H}_A$ and $|\psi_B\rangle$ is a state in $\mathcal{H}_B$, then the state of the combined system is written as:

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

The tensor product allows for the creation of entangled states, which are not separable into individual states of A and B . These entangled states are crucial in quantum computation and quantum information theory.

3.4.1 The CNOT Gate

The Controlled-NOT (CNOT) gate is a two-qubit quantum gate acting on a *control* qubit c and a *target* qubit t . It flips the target qubit if and only if the control qubit is in state $|1\rangle$, and does nothing otherwise.

In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, its matrix representation is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Exercise 3.6. *Convince yourself that the matrix CNOT changes the second qubit if the first one is set at 1.*

The construction of CNOT is equivalent to the expression

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes \sigma_X,$$

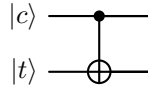
where σ_X is the Pauli-X (NOT) operator.

In an n -qubit system, the CNOT acting on qubits c and t is extended by tensoring identity operators on all other qubits:

$$U = I \otimes \cdots \otimes \text{CNOT} \otimes \cdots \otimes I.$$

However, if the control and target qubits are not adjacent, the operator must be embedded carefully, e.g., via qubit reordering or SWAP operations.

The circuit below represents a controlled-NOT (CNOT) operation acting on two qubits, arranged as horizontal lines evolving from left to right in time.



The upper wire corresponds to the *control qubit*, denoted $|c\rangle$, while the lower wire represents the *target qubit*, denoted $|t\rangle$. A filled black dot is placed on the control wire, indicating that the operation is conditioned on the state of this qubit. Specifically, the gate is activated only when the control qubit is in the state $|1\rangle$.

On the target wire, a circle containing a plus sign ($\oplus$) denotes a NOT (Pauli-X) operation, which flips the state of the qubit:

$$|0\rangle \leftrightarrow |1\rangle.$$

A vertical line connects the control dot to the target symbol, indicating that the NOT operation is applied conditionally.

The overall action of the CNOT gate is therefore:

$$|c, t\rangle \longrightarrow |c, t \oplus c\rangle,$$

where $\oplus$ denotes addition modulo 2. In words, the target qubit is flipped if and only if the control qubit is in the state $|1\rangle$; otherwise, the system remains unchanged.

The CNOT gate is fundamental for entanglement generation and, together with single-qubit gates, forms a universal set for quantum computation.

3.5 Quantum Entanglement

Quantum entanglement is a phenomenon where two or more qubits become linked, so that the state of one qubit instantly affects the state of the other, no matter how far apart they are. Entangled qubits cannot be described independently. The combined state contains more information than the individual qubits.

The classic example is the Bell state is:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (25)$$

whose measurement outcomes are perfectly correlated. It should be easy for the reader to see that if the first qubit of $|\Phi^+\rangle$ is measured as 0, then the second qubit is 0. Similarly, if the first 1 then the second is 1. The first qubit has 50% probability of being 0, but if it is so is the second bit, as the probability of the first qubit of being 0 and the second one 1 is 0% and similarly for the case when the

first qubit is 1. Hence the two states are entangled: If we measure one qubit we automatically now what the second one is going to be.

We now show how to create entangled states. We start with two qubits in $|00\rangle = |0\rangle \otimes |0\rangle$ and apply the Hadamard gate to the first qubit, that is

$$H|0\rangle \otimes |0\rangle = |+\rangle \otimes |0\rangle = 1/\sqrt{2}(|0\rangle + |1\rangle) \otimes |0\rangle = 1/\sqrt{2}(|00\rangle + |10\rangle).$$

Then we apply a CNOT gate with the first qubit control and the second qubit as target

$$\text{CNOT} \left(\frac{|00\rangle + |10\rangle}{\sqrt{2}} \right) = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

In summary, the construction is given by

$$\text{CNOT}(H|0\rangle \otimes |0\rangle)$$

Notice, we are secretly starting to do categorical quantum mechanics here [2].

We now formalise precisely the notion of entanglement by showing how the Bell state is measured. To measure the first qubit, we apply projectors on the first subsystem:

$$P_0 = |0\rangle\langle 0| \otimes I, \quad P_1 = |1\rangle\langle 1| \otimes I.$$

Both of these measurements are hermitian and unitary, hence $P_0^\dagger P_0 = P_0 P_0 = P_0^2 = P_0$ and similarly for P_1 . The probability that the first qubit is 0 is given by

$$p(0) = \langle \Phi^+ | P_0^\dagger P_0 | \Phi^+ \rangle = \frac{1}{2}.$$

whereas the resulting state in case 0 is measured is given by the stated

$$P_0|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle.$$

which, after the post-measurement normalisation becomes $|00\rangle$. Hence measuring 0 for the first qubit results in the second qubit to be 0 as well.

Exercise 3.7. *Convince yourself that this is the same when measuring 1*

3.6 Coherence and Density Matrices

In quantum computing, *coherence* refers to the ability of a quantum system to maintain well-defined phase relationships between the components of a superposition. Coherence is responsible for many uniquely quantum phenomena, including interference and entanglement. Consider a single qubit in the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.

This state is a superposition of the computational basis states $|0\rangle$ and $|1\rangle$. The relative phase between α and β carries physical significance. If this phase relationship is preserved, the system is said to be *coherent*.

Coherence allows quantum states to produce interference effects. For example, many quantum algorithms rely on constructive and destructive interference between amplitudes in order to amplify correct computational paths.

While pure states can be described using state vectors, many physical systems interact with their environment and cannot be described by a single wavefunction. In such cases we use the *density matrix* formalism.

For a pure state $|\psi\rangle$, the density matrix is defined as

$$\rho = |\psi\rangle\langle\psi|.$$

For the qubit state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

the density matrix becomes

$$\rho = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

In particular, the *diagonal elements* represent the probabilities of measuring the system in each basis state and the *off-diagonal elements* encode the phase relationships between components of the superposition. In the previous example, the nonzero off-diagonal terms

$$\rho_{01} = \rho_{10} = \frac{1}{2}$$

indicate that the system possesses coherence between the states $|0\rangle$ and $|1\rangle$. Hence the density matrix provides a convenient way to identify the presence of quantum coherence.

Now consider a situation where the system is not in a coherent superposition, but instead in a classical mixture: with probability $1/2$ the system is in $|0\rangle$ and with probability $1/2$ the system is in $|1\rangle$. The density matrix describing this mixture is

$$\rho = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Here, the off-diagonal elements vanish. This indicates that the phase relationship between the states has been lost, and therefore the system no longer exhibits quantum coherence.

In practical quantum systems, interaction with the environment gradually destroys coherence. This process is known as *decoherence*. Mathematically, decoherence suppresses the off-diagonal elements of the density matrix over time:

$$\rho = \begin{pmatrix} p_0 & c(t) \\ c^*(t) & p_1 \end{pmatrix} \longrightarrow \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix}.$$

As the coherence terms $c(t)$ approach zero, the system behaves more like a classical probabilistic system.

It is important to note that coherence is *basis dependent*. A density matrix that is diagonal in one basis may contain off-diagonal terms in another basis. Therefore, statements about coherence must always specify the basis with respect to which the density matrix is written.

Maintaining coherence is one of the central challenges in the physical realization of quantum computers. In quantum mechanics, a *state* can either be *pure* or *mixed*. A *pure state* is described by a single vector in a Hilbert space. It represents a quantum system that is in a definite state. In the case of a composite system, a pure state can be written as a *product state* or *tensor state* or an *entangled state*.

The density matrix for a pure state has the following properties:

- It is *idempotent*: $\rho^2 = \rho$.
- It has *trace 1*: $\text{trace}(\rho) = 1$, ensuring that the total probability is normalized.
- It is a *rank-1* matrix if the state is pure, meaning it can be written as a single outer product.

A *mixed state* represents a quantum system that is in a statistical mixture of pure states, which means the system's state is not known exactly. Mixed states arise when there is some uncertainty or lack of information about the system, often due to decoherence or interaction with an environment.

Mathematically, a mixed state is described by a *density matrix* ρ , which is a weighted sum of the outer products of pure states:

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

where p_i are probabilities and $|\psi_i\rangle$ are pure states. The density matrix for a mixed state satisfies $\text{trace}(\rho) = 1$ and ρ is Hermitian, that is $\rho^\dagger = \rho$.

Hence, the *density operator* (or *density matrix*) is a more general representation of a quantum state, used to describe both *pure* and *mixed* states. It is especially useful when dealing with situations where the system is in a statistical mixture of states or when the system is entangled with an environment, which is common in quantum computation.

For example, the expectation value of an observable $\hat{O}$ for a quantum state ρ is given by:

$$\langle \hat{O} \rangle = \text{trace}(\rho \hat{O})$$

This is a generalization of the expectation value for pure states, where $\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle$.

A *mixed state* represents a probabilistic mixture of pure states. If the system is in one of several pure states $|\psi_i\rangle$ with probability p_i , the density operator for the mixed state is:

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

where p_i are the probabilities that the system is in the pure state $|\psi_i\rangle$. Note that, the density matrix for a mixed state is *not idempotent* (i.e., $\rho^2 \neq \rho$), it has *more than one non-zero eigenvalue* and may have a rank greater than 1.

The density matrix for a mixed state has the following properties:

- It is *positive semi-definite*: $\langle \phi | \rho | \phi \rangle \geq 0$ for any state $|\phi\rangle$.
- It has *trace 1*: $\text{Tr}(\rho) = 1$

3.7 The Schrödinger Equation (*)

Postulate 2' *The time evolution of the state of a closed quantum system is described by the Schrödinger equation*

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = H |\psi(t)\rangle \quad (26)$$

where $\hbar$ is the Planck's constant and H is an Hermitian operator called the *Hamiltonian* of the closed system.

The Schrödinger Equation essentially postulates that the evolution in time of a quantum system is given by an Hermitian operator. To see why this is an equivalent version of Postulate 2 we have to solve the differential equation given by Schrödinger. The solution is give by

$$|\psi(t)\rangle = e^{-\frac{itH}{\hbar}} |\psi(0)\rangle. \quad (27)$$

We are going to prove that this is indeed a solution to Equation 26, but first notice that if we set $U(t) = e^{-itH/\hbar}$ then $|\psi(t)\rangle = U(t) |\psi(0)\rangle$ which is a more direct way to calculate the evolution of a quantum state at the time t starting from an initial quantum state $|\psi(0)\rangle$. If H is Hermitian, namely $H^\dagger = H$ by definition of Hermitian matrices, then $U(t)$ is unitary. To see this it suffices to compute

$$U^\dagger(t)U(t) = I$$

This is easy by noticing that $U^\dagger(t) = (e^{-iHt/\hbar})^\dagger = e^{iHt/\hbar}$, then $e^{iHt/\hbar}e^{-iHt/\hbar} = I$.

We now prove that (27) is a solution to Equation 26.

Proof of 26. The first step is to calculate the derivative of $|\psi(t)\rangle$:

$$\frac{d|\psi(t)\rangle}{dt} = \frac{-iH}{\hbar} \cdot e^{-\frac{itH}{\hbar}} |\psi(0)\rangle. \quad (28)$$

By substituting this derivative into Equation 26 we get

$$i\hbar \frac{-iH}{\hbar} \cdot e^{-\frac{itH}{\hbar}} |\psi(0)\rangle = H |\psi\rangle.$$

This can be simplified to

$$H \cdot \underbrace{e^{-\frac{itH}{\hbar}} |\psi(0)\rangle}_{|\psi(t)\rangle} = H |\psi\rangle.$$

Now notice that $e^{-\frac{itH}{\hbar}} |\psi(0)\rangle$ is $|\psi(t)\rangle$, hence the equation above can be rewritten as

$$H \cdot |\psi(t)\rangle = H |\psi\rangle$$

as wanted. □

3.8 The Bloch Sphere (*)

Consider qubit written as a linear combination of the computational basis

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Although α and β are complex numbers, the state can be expressed using two real parameters:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

The reader who is not familiar with this conversion is encouraged to take a look at the appendix, Section A.2, for a more detailed explanation of how this works.

The Bloch sphere is a way of representing qubits graphically. In Figure 14 a vector corresponds to a point on a unit sphere. The north pole is $|0\rangle$, the south pole is $|1\rangle$ and any other point is a vector in superposition. Thus, every pure qubit state is a point on the surface of the sphere whereas mixed states lie inside the sphere.

References

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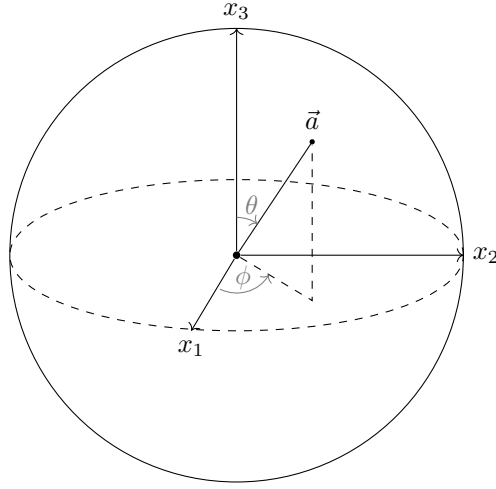


Figure 14: Bloch Sphere

A Elements of Set Theory

A *set* (or class) is an unordered collection of objects called *elements* of the set. We write $a \in X$ when a is an element of the set X and we read it as “ a belongs to X ”. Here are some important sets:

- $\emptyset$ the empty set with no elements. The reader should ponder about the difference between the empty set $\emptyset$ and the singleton set $\{\emptyset\}$ containing the empty set.
- $\mathbb{N}$ the set of *natural numbers* $\{1, 2, 3, \dots\}$
- $\mathbb{N}_0$ the set of *natural numbers with zero* $\{0, 1, 2, 3, \dots\}$
- $A \times B$ the *cartesian product* of sets A and B is the set containing the pairs (a, b) such that $a \in A$ and $b \in B$.
- $A \cup B$ the *union* of sets A and B is the set containing all elements of A and of B . If there are equal elements in A and B , these become squashed together.
- $A \uplus B$ the *disjoint union* of sets A and B is the set containing all elements $(1, a)$ for $a \in A$ and $(2, b)$ for $b \in B$.

A *relation* $R \subseteq A \times B$ is a subset of the cartesian product $A \times B$ relating some elements in A with some elements in B . A *function* is a relation $f \subseteq A \times B$ such that for an $a \in A$, there exists only one $b \in B$. We write the set of functions between A and B as $A \rightarrow B$.

There is only one function from the empty set $\emptyset$ into any other set A , that is the empty relation denoted by $! \subseteq \emptyset \times A$. Dually, there is only one function from any set A to the singleton set $\{*\}$ for an element $*$. That is the function sending every $a \in A$ into $*$. We denote this map $!$ as well, although it should be clear from the context which one we mean.

An important notion in set theory is the one of *size*, which indicates the *cardinality* of a set. Two sets are said to be *isomorphic* when they have the same cardinality. An isomorphism is given by a function $f : A \rightarrow B$ and its *inverse* $f^{-1} : B \rightarrow A$ such that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

A set A is *finite* if and only if it is isomorphic to the set $\{m \in \mathbb{N} \mid m \leq n\}$ for some $n \in \mathbb{N}$. If this is the case, it means we can enumerate the elements of A and write A as $\{a_1, a_2, a_3, \dots, a_n\}$. This is because intuitively we could write down its elements on a piece of paper in a finite amount of time.

We say a set is *infinite* if and only if it is not finite. A set is *countable* if it is isomorphic to the natural numbers.

For a set I , we denote the family of sets indexed by I as $\{A_i\}_{i \in I}$. In the case that I is finite, the family $\{A_i\}_{i \in I}$ is finite, and we can write both the finite union and product of these sets as follows:

$$A_1 \cup A_2 \cup \cdots \cup A_n$$

and similarly for the product. In the case that I is infinite, the union of the infinite family of sets can be written simply as:

$$\bigcup_{i \in I} A_i = \{a \in A_i \mid i \in I\}$$

Dually, the *infinite product* or *dependent product* is defined by the set of functions that, given an index $i \in I$, return an element in A_i :

$$\prod_{i \in I} A_i = \{f : I \rightarrow \bigcup_{i \in I} A_i \mid f(i) \in A_i\}$$

A.1 Basic Trigonometry

The sine ($\sin$) and cosine ($\cos$) functions are fundamental trigonometric functions. The unit circle provides a geometric interpretation of $\sin$ and $\cos$. Figure 15 illustrates the unit circle with an angle θ , where:

$$\cos(\theta) = x \text{ (horizontal projection), } \sin(\theta) = y \text{ (vertical projection).}$$

The *domain* of $\sin(x)$ and $\cos(x)$ is all real numbers: $x \in \mathbb{R}$ while the range of both functions is $[-1, 1]$. Moreover, both functions are periodic with a period of 2π :

$$\sin(x + 2\pi) = \sin(x), \quad \cos(x + 2\pi) = \cos(x).$$

this means that a periodic point in time (the multiples of 2π the functions go back to their starting value.

The function $\sin(x)$ is an odd function while the $\cos(x)$ function is even, meaning that :

$$\sin(-x) = -\sin(x) \quad \cos(-x) = \cos(x).$$

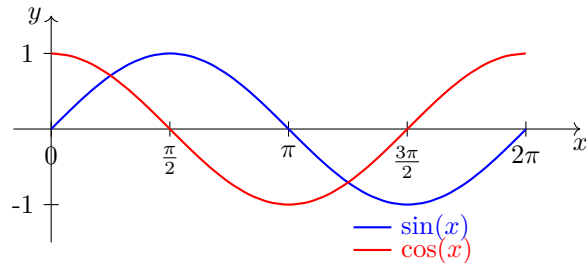
Proposition A.1 (Pythagora's identity). *The sine and cosine functions satisfy the fundamental Pythagorean identity:*

$$\sin^2(\theta) + \cos^2(\theta) = 1.$$

where θ is an angle.

We list below the values for particular key angles:

$$\begin{aligned} \sin(0) = 0, \quad \sin\left(\frac{\pi}{2}\right) = 1, \quad \sin(\pi) = 0, \quad \sin\left(\frac{3\pi}{2}\right) = -1, \quad \sin(2\pi) = 0, \\ \cos(0) = 1, \quad \cos\left(\frac{\pi}{2}\right) = 0, \quad \cos(\pi) = -1, \quad \cos\left(\frac{3\pi}{2}\right) = 0, \quad \cos(2\pi) = 1. \end{aligned}$$



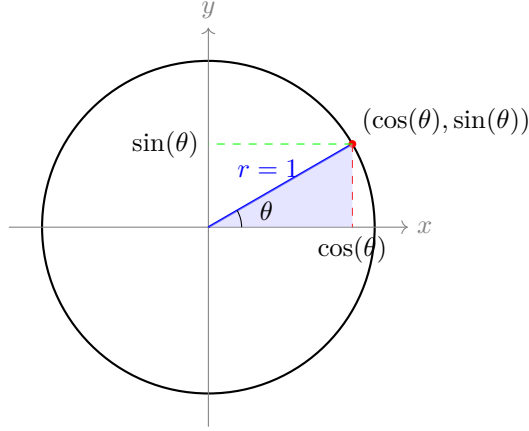


Figure 15: The Unit Circle

Finally, the sine and cosine functions satisfy the following addition formulas:

$$\sin(a + b) = \sin(a) \cos(b) + \cos(a) \sin(b),$$

$$\cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b).$$

Step 3: Parametrize the Magnitudes with θ

After factoring out the global phase, the qubit can be written as

$$|\psi\rangle = |\alpha| |0\rangle + |\beta| e^{i\phi} |1\rangle,$$

with $|\alpha|, |\beta| \geq 0$ and $|\alpha|^2 + |\beta|^2 = 1$.

Introducing the Bloch Sphere Angle θ : To map this qubit onto the Bloch sphere, we parametrize the magnitudes using a single angle $\theta \in [0, \pi]$:

$$|\alpha| = \cos \frac{\theta}{2}, \quad |\beta| = \sin \frac{\theta}{2}.$$

Why $\theta/2$?

- Using $\theta/2$ ensures the state remains normalized automatically:

$$|\alpha|^2 + |\beta|^2 = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} = 1.$$

- It also aligns the polar angle θ with the standard Bloch sphere representation:
 - $\theta = 0 \implies |\psi\rangle = |0\rangle$ (north pole)
 - $\theta = \pi \implies |\psi\rangle = |1\rangle$ (south pole)
 - $0 < \theta < \pi \implies$ superposition between $|0\rangle$ and $|1\rangle$

Intuition: This is analogous to a point on the unit circle in 2D, where

$$x^2 + y^2 = 1 \implies x = \cos \theta, y = \sin \theta.$$

Here, $|\alpha|$ and $|\beta|$ are like x and y , and the factor of $1/2$ is a conventional choice to match the Bloch sphere representation.

A.2 From Complex to Real Linear Combinations

Often it is useful to consider a vector with complex coefficients and convert into a vector with real coefficients. The key idea behind this transformation is that a complex number corresponds to a 2D real vector:

$$a + ib \leftrightarrow (a, b)$$

thus there is an isomorphism of vector spaces

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

In words, a vector of n complex entries can be represented as a vector of $2n$ entries where the first n are the real parts of the entries and the second half are the imaginary parts.

More precisely, let $|\psi\rangle$ be a vector written as a linear combination:

$$|\psi\rangle = \sum_k c_k |k\rangle, \quad c_k \in \mathbb{C}$$

Each coefficient can be decomposed into real and imaginary parts:

$$c_k = a_k + ib_k, \quad a_k, b_k \in \mathbb{R}$$

Substituting:

$$|\psi\rangle = \sum_k (a_k + ib_k) |k\rangle = \sum_k a_k |k\rangle + i \sum_k b_k |k\rangle$$

Thus the vector separates into a real and an imaginary component. We can represent the complex vector as a real vector of dimension $2n$:

$$|\psi\rangle \longleftrightarrow \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \\ b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \in \mathbb{R}^{2n}$$

The first n entries are the real parts, and the last n entries are the imaginary parts. Each basis vector $|k\rangle$ corresponds to two real directions:

$$|k\rangle \mapsto (|k\rangle_{\text{Re}}, |k\rangle_{\text{Im}})$$

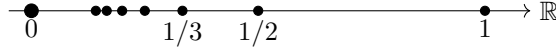


Figure 16: The Cauchy sequence $a_n = 1/n$

Example Consider the vector

$$|\psi\rangle = (1 + 2i)|0\rangle + (3 - i)|1\rangle,$$

then,

$$c_0 = 1 + 2i \rightarrow (1, 2), \quad c_1 = 3 - i \rightarrow (3, -1)$$

This the real representation is:

$$\begin{pmatrix} 1 \\ 3 \\ 2 \\ -1 \end{pmatrix}$$

Note that although the vector now has only real entries the multiplication by i becomes a rotation between real and imaginary parts and linear operators become real matrices of size $2n \times 2n$.

A.3 Notions of Metric Spaces

A.3.1 Cauchy complete spaces

A sequence (a_n) is called a *Cauchy sequence* if its terms become arbitrarily close to each other as the sequence progresses.

Definition A.1 (Cauchy sequence). *A sequence of points a_i in $\mathbb{R}^n$ is called a Cauchy sequence, iff for every $\varepsilon > 0$, there exists an index N such that for all $m, n \geq N$, the distance between the terms satisfies*

$$|a_m - a_n| < \varepsilon$$

This definition does not mention a limit; instead, it focuses only on how close the terms are to one another.

For example, the sequence $a_n = \frac{1}{n}$ depicted in Figure 16 is Cauchy in $\mathbb{R}$. As n becomes large, the values $1/n$ become very small, and the difference between any two large terms becomes arbitrarily small.

In the real numbers every Cauchy sequence converges, which is why $\mathbb{R}$ is called a complete space.

Definition A.2 (Limit). *Let (a_n) be a sequence in $\mathbb{R}$. We say that (a_n) converges to a real number L , and write*

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L,$$

if for every $\varepsilon > 0$ there exists a natural number N such that for all $n \geq N$,

$$|a_n - L| < \varepsilon.$$

A space is complete iff every Cauchy sequence has a limit.

This definition means that the terms of the sequence eventually get arbitrarily close to L . No matter how small a distance ε we choose, there is some index N after which all terms of the sequence lie within the ε -neighborhood of L . In other words, beyond a certain point, the sequence stays as close to L as we wish.

So far we have defined the notion of Cauchy sequence and limit using the space $\mathbb{R}^n$ in which the notion of distance between two points (vectors) is given by the norm of the difference between the

vectors. The norm is in turn defined using the inner product, so in order to generalise these definitions we just need to postulate a space with an inner product in which all Cauchy sequences are complete. This idea leads to the definition of a Hilbert space.

Definition A.3 (Hilbert Space). *A Hilbert space is a vector space H equipped with an inner product such that the metric induced by the norm is complete, i.e. every Cauchy sequence converges.*